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1 0. Course Information

1.1 o Course Syllabus

1.1.1 Coaching Actuaries VEE Mathematical Statistics Syllabus

To ensure course success, read the syllabus and its instructions completely. Failure to abide by course requirements may result in course failure.

Course Instructor & Coaches * Mike Carmody, FSA, CFA, MAAA (Instructor) * Chin Yew Kok, FSA, CFA (Coach) * Jonathan Lai, M.S. (Coach)

Students can ask questions on the VEE discussion forum or email a coach (Coach Chin: ckok@coachingactuaries.com). A coach will respond to student emails Monday-Friday.

Course Objectives & Components As a prerequisite, knowledge of Exam P topics is expected. The course addresses statistical concepts developed from a probabilistic framework. It introduces methods that aim to provide statistical inference regarding aspects of a population via sampling. These methods include point estimation, interval estimation, and hypothesis testing.

Due to the 120-day self-paced nature of this course, we must be stringent with our requirements:

- The course begins at the time of purchase. Assignments and the final exam must be completed within 120 days unless a course is put on hold for a \$25 fee (contact support@coachingactuaries.com).
 - If you purchased a VEE Bundle and you email us immediately to put one course on hold, the \$25 fee will be waived.
- You are required to complete all assignments **before** the final exam.
- The final exam must be scheduled and proctored. See “Proctor for Final Exam” below.

Course Grade Composition The course grade is composed of the following:

- 4 graded assignments: 50% of your final grade

Assignment	Score
Assignment 1	8% (8 questions)
Assignment 2	15% (10 questions)
Assignment 3	12% (8 questions)
Assignment 4	15% (10 questions)

- Once you begin an assignment, you will have 24 hours on a timer to complete it (you have the ability to pause and resume the timer as you would like).
 - You have **two attempts** per assignment and the **higher score of the two attempts** will be weighted toward your final grade.
 - For extra practice, you can take each assignment as many times as you like (submissions beyond the second attempt do not count toward your final grade).
- One 2-hour proctored final exam, 20 questions (each with 5 answer choices): 50%

Letter grades are: A: 85 - 100%, B: 75 - 84%. **Students need at least a B to pass the course.**

Coaching Actuaries (CA) emails grade slips to candidates and SOA on the 1st and 15th of every month (or the next business day). If you pass the course, CA will submit your name, email address and grade to SOA directly. You need to apply for VEE credit with SOA or CAS.

Applying for Credit:

- **SOA** : Learn how to apply for credit here .
 - CA will submit a list of passing candidates to the SOA.
 - The SOA will store your information until you apply for transcript credit.
- **CAS** : Learn how to apply for credit here .
 - You will need to upload to the CAS website a copy of your grade slips (both Micro and Macroeconomics or both Accounting and Finance).
 - Print this email as a pdf or take a clear and complete screenshot of this email. Upload the pdf or the image file on the CAS website.

– The Coaching Actuaries School Code is 9930.

Course Structure * The course is partitioned into suggested weekly units, but you can complete the course at your own pace. For each unit, read the online manual, watch the instructional video(s), and complete the assignments. To prepare for the assignments, solve the examples in the online manual on your own. * Adapt generates assignments and the final exam, which recreates the CBT testing style of SOA **multiple-choice** exams.

Chapter Sequence This course follows the sequence as outlined below:

Unit	Topic
1	1.1 - Random Sampling 1.2 - Sampling Distributions Assignment 1
2	2.1 - Moment and Percentile Matching Estimations 2.2 - Maximum Likelihood Estimation 2.3 - Estimators Assignment 2
3	3.1 - Confidence Intervals for Means 3.2 - Confidence Intervals for Other Parameters Assignment 3
4	4.1 - Intro to Hypothesis Testing 4.2 - Common Tests Assignment 4
6	Final Exam (covers all units above)

Learning Objectives This course covers the following learning objectives:

- Explain the concepts of random sampling, statistical inference and sampling distribution, and state and use basic sampling distributions.
- Describe the main methods of estimation and the main properties of estimators, and apply them. Methods include matching moments, percentile matching, and maximum likelihood, and properties include bias, variance, mean squared error, consistency, efficiency, and UMVUE.
- Construct confidence intervals for unknown parameters, including the mean, differences of two means, variances, and proportions.
- Test hypotheses. Concepts to be covered include Neyman-Pearson lemma, significance and power, likelihood ratio test, and information criteria. Tests should include for mean, variance, contingency tables, and goodness-of-fit.

Before the final exam, complete all assignments. Review the assignments and watch the corresponding videos or read the corresponding sections of the manual for the topics that you are weak in. **The course does not provide additional practice problems beyond the ones in the assignments.**

Proctor for Final Exam and Final Exam Details Schedule your final exam **1-2 weeks** in advance. You have two options to select a proctor:

1. Free online proctoring by a CA proctor. Click on the Reports icon to schedule your final.
 - CA offers proctoring services via Zoom from **8:30 am-4:00 pm Central Time** on Mondays-Fridays. State your preferred date and time in your email.
 - Saturday hours are **8:00 am-12:00 pm Central Time** for a small \$25 weekend fee.
 - If you need to reschedule, you will be charged a fee of \$25.
2. In-person proctoring by your chosen proctor
 - Schedule the exam **by emailing your proctor's name and work email address to proctoring@coachingactuaries.com**. The exam link will be provided to the proctor.
 - Your proctor must be an FSA (in good standing). If an FSA is not available, other acceptable designations would be: FCAS, FCIA, FFA, FIA, FIAA, FSPA, CFA, ASA, ACAS. If no Fellow/Associate is available, a high-ranking member of management who is not able to sit for exams, such as the Vice

President of Human Resources, or a university professor may also serve. Your proctor cannot be related to you.

Exam Procedures 1. The exam is closed-notes. One or two SOA-approved calculators, a writing device, and blank scratch paper are allowed. 2. Prior to the exam, your proctor will check:

- Photo identification with name printed in Latin alphabet letters (passport, driver's license, or identification card)
 - Calculator is cleared of memory, scratch paper is blank, workspace is clear of notes/books
3. Your proctor will provide the exam link; the exam will generate once the link is accessed.
4. When you have finished the exam, click "Finish Exam." Your final exam results will be displayed. The final exam cannot be accessed after the proctoring session ends.

If you **do not** pass the course, you can use the one-time Pass Guarantee that comes with your purchase. You will need to redo the course assignments and retake the final exam. Please contact support@coachingactuaries.com for assistance in this process.

1.2 Introduction

We proudly present to you the Coaching Actuaries VEE Mathematical Stats course!

This course focuses on expanding concepts from Exam P into the realm of statistics. We discuss approaches that statisticians would take to make probability applicable and useful. Our hope with this manual is that you will enjoy venturing into real-life problems with a technical outlook, in the same measure as we enjoyed authoring it.

The course is structured into the following sections:

- **Statistics and Sampling Distributions**
 - What is the foundation of statistics? How are sampling distributions crucial for statistical analysis?
- **Statistical Estimation**
 - How do we estimate unknown population parameters using sample data?
- **Confidence Intervals**
 - What are confidence intervals? How do they help us quantify the uncertainty in our statistical estimates?
- **Hypothesis Testing**
 - What is the process of hypothesis testing? How does it allow us to draw meaningful conclusions from data?

Let's begin!

2 1. Probability Distributions

2.1 1.1.1 Statistical Experiment

Suppose there is a disease spreading in the USA that has infected many people. A cure is needed for all who are infected, but practically, not all can be examined. How can a cure be discovered without examining every infected person?

A statistical experiment can be used to learn more about the disease in search for a cure.

A *statistical experiment* is an experiment conducted on a sample taken from a population. By measuring and analyzing observable variables from the sample, scientists can make *statistical inferences* about certain characteristics of the target population. The two main methods used to derive statistical inference are **estimation** and **hypothesis testing**. We will learn more about them later in the course.

2.2 1.1.2 Simple Random Sampling

A sample that poorly represents the population of interest runs a large risk of producing poor inferences. As a result, the *method of sampling* is an important process in the design of a statistical experiment.

2.2.1 Simple Random Sampling

One method of sampling is known as *simple random sampling*. Formally, let's define N as the population size, and n as the sample size. Simple random sampling occurs when all possible samples have an equal probability to be chosen from the population. Specifically, that probability is

$$\frac{1}{\binom{N}{n}}$$

Consider the following illustration:

From a classroom of 20 students, a teacher chooses 5 students to be the sample for an experiment. From combinatorics, there is a total of $\binom{20}{5} = 15,504$ different ways to pick the sample (i.e. 15,504 different sets of 5 students). A simple random sampling of 5 students is achieved if each way has a $\frac{1}{15,504}$ chance of being the sample.

Example 1.1.1 For a behavioral study, a psychologist has a list of 48 volunteers. Each volunteer is assigned a number from 1 to 48.

In forming a group of four with simple random sampling, what is the probability that all four were assigned an even number?

Solution There are

$$\binom{48}{4} = 194,580$$

possible ways to select four volunteers from 48 of them. Under a simple random sample, all possible groups have the same probability, namely $\frac{1}{194,580}$, to be the chosen group.

There are $48 \div 2 = 24$ even numbers from integers 1 to 48. The number of groups that have four even numbers is

$$\binom{24}{4} = 10,626$$

Since each of the 10,626 groups has a $\frac{1}{194,580}$ probability, the answer is

$$\frac{10,626}{194,580} = 0.0546$$

[sq]

2.3 1.2.1 Normal Distribution

Start with a random sample $X_1, X_2, \dots, X_n$ (i.e. identically and independently distributed random variables) with a known common mean μ , and a known common variance σ^2 . The *sample mean*, $\bar{X}$, is the arithmetic mean of the random sample.

$$\bar{X} = \frac{\sum_{i=1}^n X_i}{n}$$

The mean of $\bar{X}$ is μ , and the variance of $\bar{X}$ is $\frac{\sigma^2}{n}$. In addition, $\bar{X}$ is a statistic, since it takes the average of n variables. Its sampling distribution is

- a normal distribution if the X 's are normally distributed, or
- an approximate normal distribution if the Central Limit Theorem applies.

Coach's Remarks Here is a refresher on the Central Limit Theorem:

For a sufficiently large random sample, $\bar{X}$ approximately follows a normal distribution, regardless of the distribution from which the sample is drawn.

The parameters of a normal distribution are its mean and variance. We denote Z as the standard normal (i.e. mean = 0, variance = 1) random variable. Specifically, for any normal random variable X ,

$$Z = \frac{X - E[X]}{\sqrt{Var[X]}}$$

Recall that probabilities from a normal distribution are calculated by first standardizing the normal random variable, then finding the needed values from a standard normal distribution table (or Z -table).

The Z -table below illustrates how $Pr(Z < 1.55) = 0.9394$. When completing an assignment or the final exam for this course, the table will be available via the Exhibit button.

Let's try an example.

Example 1.2.1 A teacher finds test scores to be normally distributed with mean 82 and standard deviation 3. She randomly selects 4 students from the class and checks their test scores.

What is the probability that the sample mean of these 4 scores will be greater than 80, but less than 85?

Solution We know the test scores follow a normal distribution. Therefore, the sample mean score, $\bar{X}$, also follows a normal distribution with mean of

$$E[\bar{X}] = \mu = 82$$

and variance of

$$Var[\bar{X}] = \frac{\sigma^2}{n} = \frac{3^2}{4} = \frac{9}{4}$$

The probability of interest is

$$Pr(80 < \bar{X} < 85)$$

NORMAL DISTRIBUTION TABLE

Entries represent the area under the standardized normal distribution from $-\infty$ to z , $\Pr(Z < z)$

The value of z to the first decimal is given in the left column. The second decimal place is given in the top row.

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998
3.5	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998
3.6	0.9998	0.9998	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.7	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.8	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.9	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

Values of z for selected values of $\Pr(Z < z)$							
z	0.842	1.036	1.282	1.645	1.960	2.326	2.576
$\Pr(Z < z)$	0.800	0.850	0.900	0.950	0.975	0.990	0.995

Figure 1: Standard normal distribution table with title NORMAL DISTRIBUTION TABLE showing cumulative probabilities from negative infinity to z , denoted $\Pr$ of Z less than z , for z -values from 0.0 to 3.9 in rows and second decimal places 0.00 to 0.09 in columns. A supplementary table titled Values of z for selected values of $\Pr$ of Z less than z lists z -values 0.842, 1.036, 1.282, 1.645, 1.960, 2.326, and 2.576 corresponding to probabilities 0.800, 0.850, 0.900, 0.950, 0.975, 0.990, and 0.995.

To solve this probability, standardize the normal random variable by subtracting its mean, then dividing by its standard deviation. Use the normal table to find the necessary probabilities.

$$\begin{aligned}
 Pr(80 < \bar{X} < 85) &= Pr\left(\frac{80-E[\bar{X}]}{\sqrt{Var[\bar{X}]}} < \frac{\bar{X}-E[\bar{X}]}{\sqrt{Var[\bar{X}]}} < \frac{85-E[\bar{X}]}{\sqrt{Var[\bar{X}]}}\right) \\
 &= Pr\left(\frac{80-82}{\sqrt{9/4}} < Z < \frac{85-82}{\sqrt{9/4}}\right) \\
 &= Pr(-1.33 < Z < 2) \\
 &= Pr(Z < 2) - Pr(Z < -1.33) \\
 &= 0.9772 - (1 - 0.9082) \\
 &= 0.8854
 \end{aligned}$$

[sq]

2.4 1.2.2 Chi Squared Distribution

Continuing with the random sample from before, let's define another statistic:

$$\frac{\sum_{i=1}^n (X_i - \bar{X})^2}{\sigma^2}$$

which is also often written as

$$\frac{(n-1)S^2}{\sigma^2}$$

where S^2 is the *unbiased sample variance*.

$$S^2 = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n-1}$$

If the X 's follow a normal distribution, then the sampling distribution of $\frac{(n-1)S^2}{\sigma^2}$ is a χ^2 (chi-squared) distribution with $n-1$ degrees of freedom. This statistic is particularly helpful for a proof we will see in the next subsection.

The χ^2 distribution is a continuous distribution with one parameter called degrees of freedom. By definition, for r independent squared standard normal random variables, their sum

$$\sum_{i=1}^r Z_i^2$$

has a χ^2 distribution with r degrees of freedom.

This means it can be proven that, for a random sample drawn from a normal distribution, $\frac{(n-1)S^2}{\sigma^2}$ has the same distribution as the sum of $n-1$ independent squared standard normal random variables.

Several applications of the χ^2 distribution will be covered in detail towards the end of this course.

To find χ^2 probabilities, it is common to use a computer software these days. But in this course, a small table of percentiles will be given in each problem when necessary. You are expected to know how to derive the answer based on the information provided.

Example 1.2.2 Heavenly Airlines is testing the efficiency of its new type of airplane. A manager believes that a flight time without refueling is normally distributed with standard deviation of 2.10525 hours. A researcher gathers a random sample of 20 airplanes to test.

Let $\chi_{p,\nu}^2$ be the 100p th percentile of a chi-squared random variable with ν degrees of freedom. The following table lists values of $\chi_{p,\nu}^2$ for specific combinations of p and ν :

	$p = 0.95$	$p = 0.99$	$p = 0.995$
$\nu = 19$	30.144	36.191	38.582
$\nu = 20$	31.410	37.566	39.997

Based on the belief of the manager, calculate the probability that the sample variance of the 20 flight times is less than 9.

Solution As a refresher, π_p is the 100p th percentile of X when $Pr(X \leq \pi_p) = p$.

The problem gives $\sigma = 2.10525$ and $n = 20$. We can also conclude that $\frac{(n-1)S^2}{\sigma^2}$ follows a χ^2 distribution with $20 - 1 = 19$ degrees of freedom.

The goal is

$$\begin{aligned}
 Pr(S^2 < 9) &= Pr\left(\frac{(n-1)S^2}{\sigma^2} < \frac{(n-1)9}{\sigma^2}\right) \\
 &= Pr\left(\frac{(n-1)S^2}{\sigma^2} < \frac{(20-1)9}{2.10525^2}\right) \\
 &= Pr\left(\frac{(n-1)S^2}{\sigma^2} < 38.582\right) \\
 &= 0.995
 \end{aligned}$$

[sq]

Coach's Remarks Unless stated otherwise, “sample variance” is understood to refer to the unbiased sample variance, S^2 .

Since a limited number of percentiles is provided in a problem, you may need to deduce an answer using inequalities. Consider the next example.

Example 1.2.3 The daily change in the WXY stock price follows a normal distribution. An analyst notes the stock price changes for a random sample of 25 days.

Let $\chi_{p,\nu}^2$ be the 100p th percentile of a chi-squared random variable with ν degrees of freedom. The following table lists values of $\chi_{p,\nu}^2$ for specific combinations of p and ν :

	$p = 0.3$	$p = 0.7$
$\nu = 24$	19.943	27.096
$\nu = 25$	20.867	28.172

Let q be the probability that the sample variance of the 25 daily changes is under 0.85 of the true variance. Determine whether q is less than 30%, between 30% and 70%, or greater than 70%.

Solution In this example, $\frac{(n-1)S^2}{\sigma^2}$ follows a χ^2 distribution with $25 - 1 = 24$ degrees of freedom. The probability of interest is

$$\begin{aligned} q &= Pr(S^2 < 0.85\sigma^2) \\ &= Pr\left(\frac{S^2}{\sigma^2} < 0.85\right) \\ &= Pr\left(\frac{(n-1)S^2}{\sigma^2} < (25-1)0.85\right) \\ &= Pr\left(\frac{(n-1)S^2}{\sigma^2} < 20.4\right) \end{aligned}$$

With 24 degrees of freedom, conclude that

$$19.943 < 20.4 < 27.096$$

$$\Rightarrow Pr\left(\frac{(n-1)S^2}{\sigma^2} < 19.943\right) < Pr\left(\frac{(n-1)S^2}{\sigma^2} < 20.4\right) < Pr\left(\frac{(n-1)S^2}{\sigma^2} < 27.096\right)$$

$$\Rightarrow 30\% < q < 70\%$$

$$[sq]$$

2.5 1.2.3 Student S T Distribution

Let's resume where we ended with statistic $\bar{X}$ having a normal sampling distribution. Recall that standardizing $\bar{X}$ gives the standard normal random variable.

$$Z = \frac{\bar{X} - E[\bar{X}]}{\sqrt{Var[\bar{X}]}} = \frac{\bar{X} - \mu}{\sqrt{\sigma^2/n}} = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

Now imagine that the value of σ^2 is not known. In its place, substitute in the sample variance, S^2 . This produces the statistic

$$\frac{\bar{X} - \mu}{S/\sqrt{n}}$$

If the X 's follow a normal distribution, then the sampling distribution of $\frac{\bar{X} - \mu}{S/\sqrt{n}}$ is a Student's t -distribution with $n - 1$ degrees of freedom.

Defining characteristics of this statistic include:

- Bypasses the need to know σ^2
- Similar to a standard normal random variable

The t -distribution is a continuous distribution with one parameter called degrees of freedom. It is very similar to a standard normal distribution; its density function is also symmetric about 0. It is primarily different from a standard normal by having a larger kurtosis (i.e. higher chance for outliers). In fact, as the degrees of freedom approaches ∞ , a t -distribution converges to a standard normal distribution.

By definition, for a standard normal random variable Z independent of a χ^2 random variable Y with ν degrees of freedom, the expression

$$\frac{Z}{\sqrt{Y/\nu}}$$

has a t -distribution with ν degrees of freedom.

More Information For a random sample drawn from a normal distribution, it can be proven that $\frac{\bar{X}-\mu}{\sigma/\sqrt{n}}$ is independent of $\frac{(n-1)S^2}{\sigma^2}$, thus showing that

$$\frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \div \sqrt{\frac{(n-1)S^2}{(n-1)\sigma^2}} = \frac{\bar{X}-\mu}{S/\sqrt{n}}$$

is t -distributed with $n-1$ degrees of freedom.

As was done for χ^2 , we will provide a small table of t percentiles in each problem when necessary.

Let's try several examples.

Example 1.2.4 A teacher finds test scores to be normally distributed with mean 82. She randomly selects 4 students from the class and checks their test scores.

Let $t_{p,\nu}$ be the 100 p th percentile of a t random variable with ν degrees of freedom. The following table lists values of $t_{p,\nu}$ for specific combinations of p and ν :

	$p = 0.8627$	$p = 0.8965$	$p = 0.9303$
$\nu = 3$	1.333	1.604	2.000
$\nu = 4$	1.265	1.504	1.842

What is the probability that the sample mean of the 4 scores minus 82 is greater than -0.802 times the sample standard deviation of the 4 scores?

Solution Since the sample size is 4, the appropriate t -distribution has $4-1=3$ degrees of freedom.

$$\begin{aligned} Pr(\bar{X} - 82 > -0.802S) &= Pr(\bar{X} - \mu > -0.802S) \\ &= Pr\left(\frac{\bar{X}-\mu}{S/\sqrt{n}} > \frac{-0.802S}{S/\sqrt{4}}\right) \\ &= Pr\left(\frac{\bar{X}-\mu}{S/\sqrt{n}} > -1.604\right) \end{aligned}$$

Like the standard normal distribution, the t -distribution is symmetric about 0. Thus,

$$\begin{aligned} Pr\left(\frac{\bar{X}-\mu}{S/\sqrt{n}} > -1.604\right) &= Pr\left(\frac{\bar{X}-\mu}{S/\sqrt{n}} < 1.604\right) \\ &= 0.8965 \end{aligned}$$

[sq]

Example 1.2.5 Heavenly Airlines is testing the efficiency of its new type of airplane. A manager believes that a flight time without refueling is normally distributed with mean μ . A researcher gathers a random sample of 20 airplanes to test.

The probability that the sample mean is within 0.468 sample standard deviations from μ is 0.95.
 Calculate the probability that the sample mean minus μ is less than 0.468 sample standard deviations.

Solution The goal is to calculate

$$\begin{aligned} Pr(\bar{X} - \mu < 0.468S) &= Pr\left(\frac{\bar{X} - \mu}{S/\sqrt{n}} < \frac{0.468S}{S/\sqrt{20}}\right) \\ &= Pr\left(\frac{\bar{X} - \mu}{S/\sqrt{n}} < 2.093\right) \end{aligned}$$

We are told that

$$\begin{aligned} Pr(\mu - 0.468S < \bar{X} < \mu + 0.468S) &= 0.95 \\ \Rightarrow Pr(-0.468S < \bar{X} - \mu < 0.468S) &= 0.95 \\ \Rightarrow Pr\left(-2.093 < \frac{\bar{X} - \mu}{S/\sqrt{n}} < 2.093\right) &= 0.95 \end{aligned}$$

Translating into a graph,

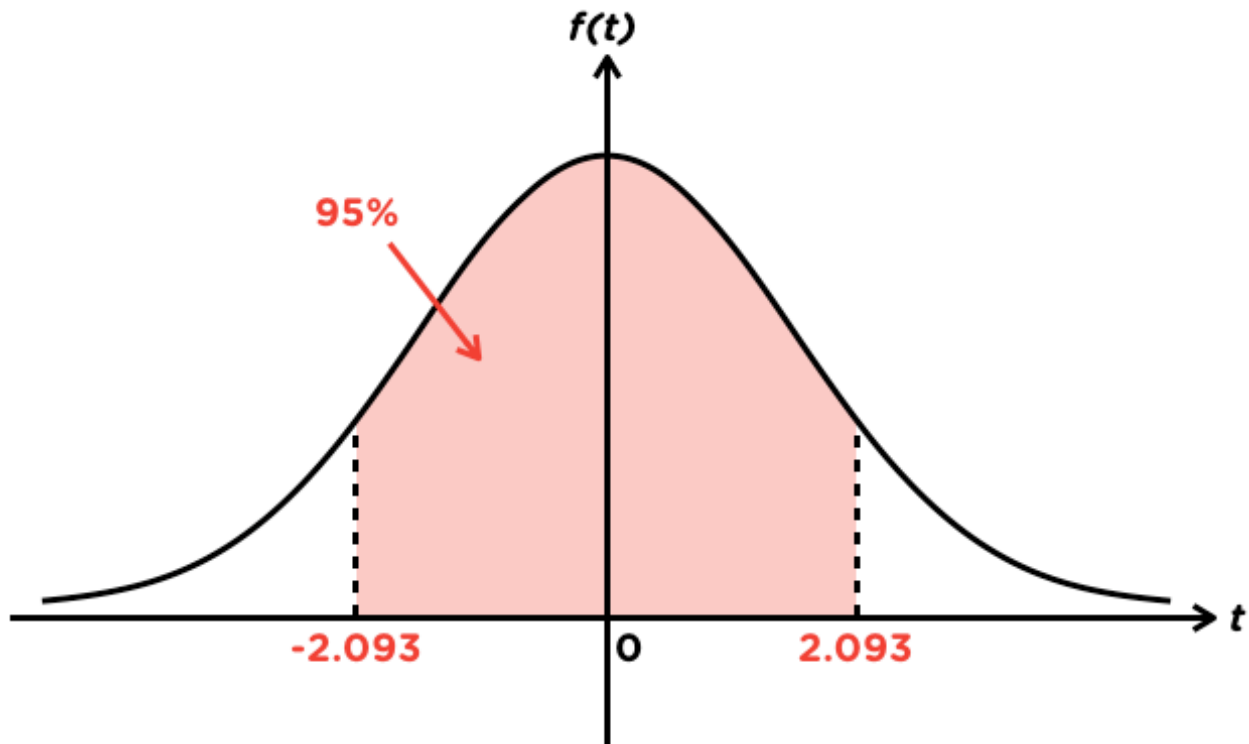


Figure 2: Probability density function graph of Student's t-distribution with horizontal axis t and vertical axis f of t . The symmetric bell curve centered at 0 has vertical dashed lines at negative 2.093 and 2.093, with the central area between them shaded and labeled 95 percent indicating the central probability region.

Hence, the probability in the tails sum to 5%. Since 5% must be equally divided between the two tails, These graphs show that the answer is

$$Pr\left(\frac{\bar{X} - \mu}{S/\sqrt{n}} < 2.093\right) = 0.025 + 0.95 = 0.975$$

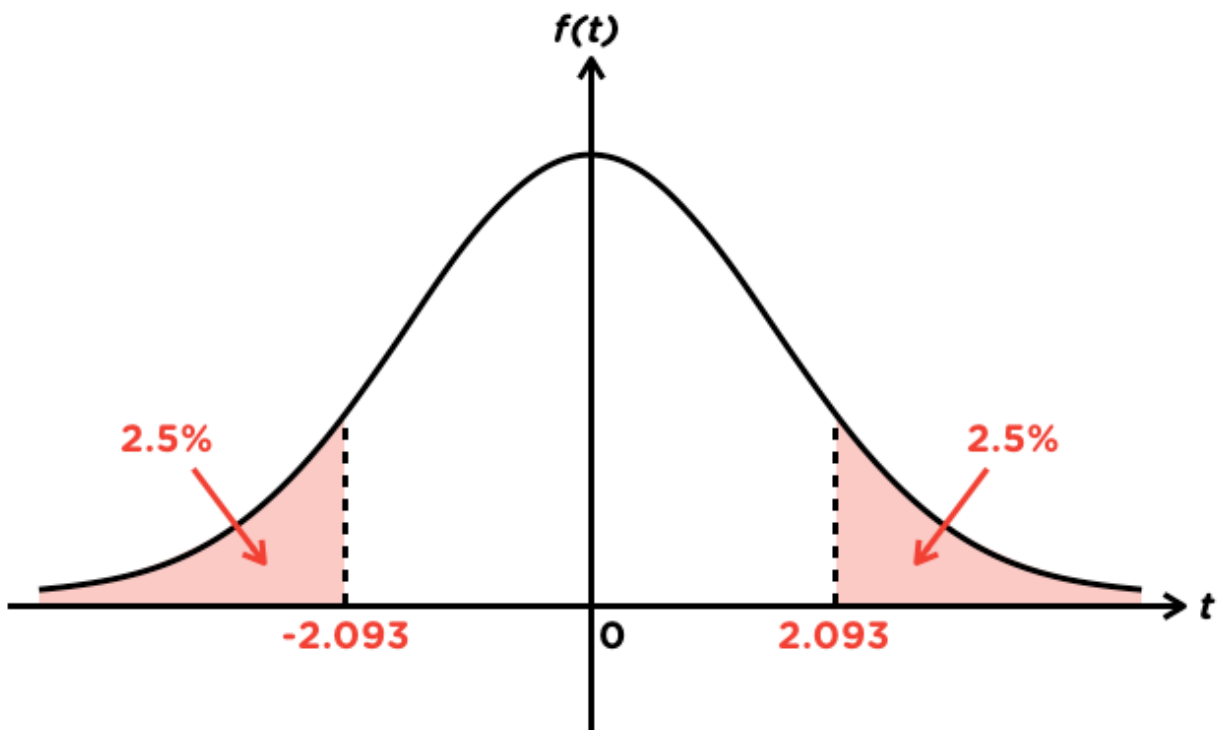


Figure 3: Line graph of a Student's t-distribution probability density function f of t versus t , symmetric about zero. Dashed vertical lines mark critical values at negative 2.093 and 2.093, with the tail regions beyond each line shaded and labeled 2.5 percent.

$$[sq]$$

Coach's Remarks We can generalize this result in math notation:

For a continuous random variable Y that is symmetric about 0,

$$\begin{aligned} Pr(-a < Y < a) = k &\Rightarrow Pr(Y < -a) = \frac{1-k}{2} \\ &\Rightarrow Pr(Y < a) = \frac{1+k}{2} \end{aligned}$$

This will be useful for the material on confidence intervals.

2.6 1.2 Overview

A *statistic* is a function of the variables observed from a sample. In probability terminology, a statistic is a random variable; the distribution of a statistic is called a *sampling distribution*. Statistical inference revolves around the use of probability to shed light on observed data from a sample.

The rest of this subsection will discuss three distributions that are foundational to statistical inference.

3 2. Point Estimation

3.1 2.1.1 Estimation

We mentioned that *estimation* is one of two ways to derive statistical inference.

As researchers study a population, they construct a model to describe reality as accurately as possible. A useful model will need specified *parameter* values that capture and quantify certain properties of the population. Examples of parameters include the population mean and population variance. However, a handful of practical issues makes it impossible to obtain most parameter values. As a result, researchers instead **estimate** the parameters based on an observed sample.

Estimation can be accomplished with *point estimation* and/or *interval estimation*. This section will focus on point estimation. The objective is to obtain a single value as our “best guess” for a parameter. But there isn’t just one way to estimate a parameter; we can apply different *methods of estimation*. In this course, we will cover three methods:

1. Method of matching moments
2. Percentile matching
3. Maximum likelihood estimation (MLE)

We begin with the first two methods in this subsection.

3.2 2.1.2 Method Of Matching Moments

For a decent-sized sample drawn from a distribution, characteristics of the observations should be similar to those of the distribution. The *method of matching moments* employs this idea to the characteristic of moments.

Let’s denote the k th (theoretical) moment of random variable X as

$$E[X^k]$$

and the k th sample moment as

$$\frac{\sum_{i=1}^n x_i^k}{n}$$

where x_i is the i th observed value from the sample of size n .

If X 's distribution has r number of parameters that require estimation, then their estimates would be the values that satisfy the following set of equations:

$$E[X^k] = \frac{\sum_{i=1}^n x_i^k}{n}, k = 1, 2, \dots, r$$

In short, this method determines the parameters that would cause the theoretical moments to equal the sample moments.

Let's illustrate the method with a couple of examples.

Example 2.1.1 The following data records the test scores of 33 students:

Number of Students	Test Score
3	60
9	70
12	80
7	90
2	100

Suppose that test scores follow an exponential distribution with mean θ .

Estimate θ using the method of matching moments.

Solution Since there is only one parameter to estimate, only one equation is needed:

$$E[X] = \frac{\sum_{i=1}^n x_i}{n}$$

We are told that the mean, $E[X]$, of the exponential distribution is simply θ .

From the data,

$$\begin{aligned} \frac{\sum_{i=1}^n x_i}{n} &= \frac{60+60+60+70+\dots+90+100+100}{33} \\ &= \frac{60(3)+70(9)+80(12)+90(7)+100(2)}{33} \\ &= 78.7879 \end{aligned}$$

Thus, the estimate of θ is

$$\hat{\theta} = 78.788$$

$$[sq]$$

Coach's Remarks The “hat” symbol will denote a generic use of point estimation. Hence, $\hat{\theta}$ here translates to “estimate of θ ”.

Example 2.1.2 The following data records the test scores of 33 students:

Number of Students	Test Score
3	60
9	70
12	80
7	90
2	100

Suppose that test scores follow a gamma distribution with mean $\alpha\theta$ and second moment $(\alpha + 1)\alpha\theta^2$. Estimate α and θ using the method of matching moments.

Solution With two parameters to estimate, two equations are needed:

$$\begin{aligned} E[X] &= \frac{\sum_{i=1}^n x_i}{n} \\ E[X^2] &= \frac{\sum_{i=1}^n x_i^2}{n} \end{aligned}$$

We are told that

$$\begin{aligned} E[X] &= \alpha\theta \\ E[X^2] &= (\alpha + 1)\alpha\theta^2 \end{aligned}$$

From the data,

$$\begin{aligned} \frac{\sum_{i=1}^n x_i}{n} &= 78.7879 \\ \frac{\sum_{i=1}^n x_i^2}{n} &= \frac{60^2(3)+70^2(9)+80^2(12)+90^2(7)+100^2(2)}{33} \\ &= 6,315.1515 \end{aligned}$$

Thus, the two equations are

$$\begin{aligned} \alpha\theta &= 78.7879 \\ (\alpha + 1)\alpha\theta^2 &= 6,315.1515 \end{aligned}$$

Rewrite the second equation in terms of $\alpha\theta$, and substitute in the value. Solve for α and θ .

$$\begin{aligned} (\alpha + 1)\alpha\theta^2 &= (\alpha\theta)^2 + (\alpha\theta)\theta \\ 6,315.1515 &= (78.7879)^2 + (78.7879)\theta \\ \theta &= \frac{6,315.1515 - (78.7879)^2}{78.7879} \\ &= 1.366 \\ \alpha &= \frac{78.7879}{1.366} \\ &= 57.679 \end{aligned}$$

The estimates are the parameter values that satisfy the equations. Thus

$$\hat{\alpha} = 57.679, \hat{\theta} = 1.366$$

$$[sq]$$

3.3 2.1.3 Percentile Matching

Aside from moments, there are many other characteristics of the observations and distribution that should align. The *percentile matching* method focuses on the characteristic of percentiles to obtain parameter estimates.

Let's denote π_p as the $100p$ th percentile of random variable X , meaning

$$Pr(X \leq \pi_p) = F(\pi_p) = p$$

In addition, denote π'_p as the $100p$ th percentile of the sample observations.

3.3.1 Basic Principle of Percentile Matching

If X 's distribution has r number of parameters that require estimation, then their estimates would be the values that satisfy the following set of equations:

$$\pi_{p_k} = \pi'_{p_k}, k = 1, 2, \dots, r$$

where the p_k 's are arbitrarily chosen. For this course, the problem will specify which percentiles should be matched.

Example 2.1.3 Assume the median of a sample is 250. Suppose the data came from an exponential distribution with CDF

$$F(x) = 1 - e^{-x/\theta}, x > 0$$

Estimate θ by matching the medians.

Solution Recall that the median is another name for the 50 th percentile. Therefore, we need to match

$$\pi_{0.5} = \pi'_{0.5} = 250$$

By definition,

$$F(\pi_{0.5}) = 1 - e^{-\pi_{0.5}/\theta} = 0.5$$

Evaluate $\pi_{0.5} = 250$, and solve for θ to calculate its estimate.

$$\begin{aligned} 1 - e^{-250/\theta} &= 0.5 \\ e^{-250/\theta} &= 1 - 0.5 \\ -\frac{250}{\theta} &= \ln(1 - 0.5) \end{aligned}$$

$$\hat{\theta} = -\frac{250}{\ln(1 - 0.5)} = 360.674$$

$$[sq]$$

In the example above, we were conveniently given that the median of the sample is 250. In reality, there are several different approaches to compute sample percentiles, each potentially producing a different answer. In this course, we will use the *smoothed empirical percentile* approach.

3.3.2 Smoothed Empirical Percentile

In this approach,

$$\pi'_p = \begin{cases} (1 - c)x_{(b)} + cx_{(b+1)}, & 1 \leq p(n + 1) \leq n \\ \text{undefined}, & \text{otherwise} \end{cases}$$

where

- n is the sample size,
- $x_{(i)}$ is the i th observation in **ascending order**,
- $b = \lfloor p(n + 1) \rfloor$, i.e. round $p(n + 1)$ **down** to the nearest integer, and
- $c = p(n + 1) - b$.

The formula above might look intimidating, but the logic is rather straightforward. Instead of memorizing the formula, consider the following steps:

1. Calculate $p(n + 1)$. For this course, you won't have to worry about it being outside the interval $[1, n]$.
 - For example, if $p = 0.65$ and $n = 34$, then $p(n + 1) = 22.75$. You may interpret this to loosely mean "the 65 th percentile of the sample is the 22.75 th observation in ascending order".
2. Calculate π'_p by linearly interpolating between $x_{(b)}$ and $x_{(b+1)}$.
 - Since 22.75 is between 22 and 23, we need to interpolate between $x_{(22)}$ and $x_{(23)}$, the 22 nd and 23 rd observations in ascending order. Take the numbers after the decimal of $p(n + 1)$ as the weight that's multiplied to the larger observation. Then, the smaller observation gets the complement weight. In other words, $\pi'_{0.65} = 0.25x_{(22)} + 0.75x_{(23)}$, where 0.75 is taken from 22.75, and $0.25 = 1 - 0.75$.
 - If $p(n + 1)$ is an integer instead, the formula simplifies to $\pi'_p = x_{(b)}$. This is consistent with the procedure above; integers have 0's after the decimal, so the larger observation will receive a weight of 0.

Let's see this in action with a few examples:

Write the expression that calculates the 34 th percentile of a sample with size 7.

$$p(n + 1) = 0.34(7 + 1) = 2.72$$

2.72 tells us to interpolate between the 2 nd and 3 rd observations in ascending order, with respective weights $1 - 0.72 = 0.28$ and 0.72 . Thus,

$$\pi'_{0.34} = 0.28x_{(2)} + 0.72x_{(3)}$$

Write the expression that calculates the 84 th percentile of a sample with size 17.

$$p(n + 1) = 0.84(17 + 1) = 15.12$$

15.12 tells us to interpolate between the 15 th and 16 th observations in ascending order, with respective weights $1 - 0.12 = 0.88$ and 0.12 . Thus,

$$\pi'_{0.84} = 0.88x_{(15)} + 0.12x_{(16)}$$

Write the expression that calculates the 40 th percentile of a sample with size 19.

$$p(n + 1) = 0.4(19 + 1) = 8$$

Since 8 is an integer, no interpolation is necessary.

$$\pi'_{0.4} = x_{(8)}$$

Coach's Remarks An alternative and perhaps more intuitive way to remember which weight goes to which observation is: $p(n + 1)$ hints at which is the **closer** observation; it gets the **larger** weight.

Revisiting the examples above,

- $p(n + 1) = 2.72$ is closer to 3 than 2. Therefore, the larger weight of 0.72 is multiplied to $x_{(3)}$, and the smaller weight of 0.28 is multiplied to $x_{(2)}$.
- $p(n + 1) = 15.12$ is closer to 15 than 16. Therefore, the larger weight of 0.88 is multiplied to $x_{(15)}$, and the smaller weight of 0.12 is multiplied to $x_{(16)}$.

Now let's apply this concept to a proper example.

Example 2.1.4 A soccer fan records the time it takes his favorite team to score one goal in 16 random matches.

15	35	60	85	33	69	88	44
35	78	90	32	2	68	23	19

Assume that the data follows a Weibull distribution with CDF

$$F(x) = 1 - e^{-(x/\theta)^\tau}, x > 0$$

Estimate θ and τ by matching the 40 th and 60 th percentiles.

Solution Our first step is to organize the data in ascending order.

2	15	19	23	32	33	35	35
44	60	68	69	78	85	88	90

Next, calculate the 40 th and 60 th sample percentiles.

For the 40 th sample percentile,

$$p(n + 1) = 0.4(16 + 1) = 6.8$$

$$\begin{aligned}
\pi'_{0.4} &= 0.2x_{(6)} + 0.8x_{(7)} \\
&= 0.2(33) + 0.8(35) \\
&= 34.6
\end{aligned}$$

For the 60 th sample percentile,

$$p(n+1) = 0.6(16+1) = 10.2$$

$$\begin{aligned}
\pi'_{0.6} &= 0.8x_{(10)} + 0.2x_{(11)} \\
&= 0.8(60) + 0.2(68) \\
&= 61.6
\end{aligned}$$

Therefore, we need to match

$$\begin{aligned}
\pi_{0.4} &= \pi'_{0.4} = 34.6 \\
\pi_{0.6} &= \pi'_{0.6} = 61.6
\end{aligned}$$

which lead to the equations

$$\begin{aligned}
1 - e^{-(34.6/\theta)^\tau} &= 0.4 \\
1 - e^{-(61.6/\theta)^\tau} &= 0.6
\end{aligned}$$

First, simplify each equation in a similar way that was shown in Example 2.1.3. The resulting equations are

$$\begin{aligned}
-\left(\frac{34.6}{\theta}\right)^\tau &= \ln(0.6) \\
-\left(\frac{61.6}{\theta}\right)^\tau &= \ln(0.4)
\end{aligned}$$

We can solve for τ by dividing the two equations. As a result,

$$\begin{aligned}
-\left(\frac{34.6}{\theta}\right)^\tau / -\left(\frac{61.6}{\theta}\right)^\tau &= \ln(0.6)/\ln(0.4) \\
\left(\frac{34.6}{61.6}\right)^\tau &= 0.5575 \\
\tau \ln\left(\frac{34.6}{61.6}\right) &= \ln(0.5575) \\
\tau &= \ln(0.5575)/\ln\left(\frac{34.6}{61.6}\right) \\
&= 1.013
\end{aligned}$$

$$\begin{aligned}
-\left(\frac{34.6}{\theta}\right)^{1.013} &= \ln(0.6) \\
\theta^{1.013} &= -\frac{34.6^{1.013}}{\ln(0.6)} \\
\theta &= \left(-\frac{34.6^{1.013}}{\ln(0.6)}\right)^{1.013^{-1}} \\
&= 67.152
\end{aligned}$$

$$\hat{\theta} = 67.152, \hat{\tau} = 1.013$$

[sq]

3.4 2.1.4 Other Comments

As illustrated by the examples, both methods are easy to perform. However, they come with two caveats:

1. The set of equations do not guarantee a unique solution of parameter estimates. In the case of having several solutions, there is no objective way to select one.
2. It is assumed the sample is drawn from **one** specific parametric distribution. Depending on the data, this assumption may not hold.

Often, these methods are considered substitutes to arguably better approaches that failed to work in a particular situation. But as is the case for point estimation in general, we should avoid basing our judgment solely on the results. It is certainly **wrong** to think that the estimates are the true parameter values; estimates merely provide an educated guess of their true counterparts.

3.5 2.2.1 Complete Data

Let's start by first understanding *likelihood*. Informally, “likelihood” is interchangeable with “probability”. The proper definition for the likelihood of an observation is a probability function evaluated at the observation. For example, if a random variable with probability function $f(x)$ is observed to have resulted in c , then the likelihood of this observation is $f(c)$.

As a result, for independently drawn “complete” observations (i.e. the specific value of each observation is known), the *likelihood function* is the product of each observation's likelihood. Assuming that $x_1, x_2, \dots, x_n$ refer to the observed values, the likelihood function is

$$\prod_{i=1}^n f(x_i)$$

Coach's Remarks Here, we are using $f(\cdot)$ to represent a probability function applicable to either discrete or continuous distributions.

In addition, the current notation assumes that the observations come from one distribution. This is not a requirement for MLE, but it will be the extent of our consideration for this course.

Let $f(x)$ involve a generic parameter θ . Keep in mind that the value of θ is not known, seeing that the goal is to estimate it. In other words, the likelihood function is a function of θ . Thus, we denote the likelihood function of θ as

$$L(\theta) = \prod_{i=1}^n f(x_i)$$

To solidify these concepts, consider this short example.

You observed the values 3, 1, and 2, believing to have independently come from a Poisson distribution with mean λ . What is the likelihood function of λ ?

The PMF of a Poisson distribution with mean λ is

$$Pr(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

Therefore, the likelihood function of λ is

$$L(\lambda) = \left(\frac{e^{-\lambda}\lambda^3}{3!}\right)\left(\frac{e^{-\lambda}\lambda^1}{1!}\right)\left(\frac{e^{-\lambda}\lambda^2}{2!}\right) \\ = \frac{e^{-3\lambda}\lambda^6}{12} \frac{e^{-3\lambda}\lambda^6}{12}$$

So how does MLE use the likelihood function to estimate the parameter of interest? The estimate is the parameter value that **maximizes** the likelihood function. In many cases, this is a just calculus problem: take the first derivative of the likelihood function, and set it to equal 0; the solution of that equation is the MLE estimate. In math notation, the estimate is the value of θ that solves

$$\frac{d}{d\theta}L(\theta) = L'(\theta) = 0$$

However, it is often the case that taking the derivative of $L(\theta)$ is tedious. A convenient alternative is to instead maximize $l(\theta) = \ln[L(\theta)]$, the *log-likelihood function*, whose derivative is likely easier to compute. We are guaranteed that maximizing the natural logarithm of any function will result in the same solution as maximizing the original function.

What is the MLE estimate of λ for the previous example?

First, calculate the log-likelihood function:

$$l(\lambda) = \ln[L(\lambda)] \\ = \ln\left[\frac{e^{-3\lambda}\lambda^6}{12}\right] \\ = -3\lambda + 6\ln[\lambda] - \ln[12]$$

To maximize the function, set its first derivative equal to 0. Solving for λ produces an estimate of 2.

$$l'(\lambda) = \frac{d}{d\lambda}l(\lambda) \\ = -3 + \frac{6}{\lambda} = 0$$

$$\hat{\lambda} = \frac{6}{3} = 2$$

Example 2.2.1 The following data records the test scores of 33 students:

Number of Students	Test Score
3	60
9	70
12	80
7	90
2	100

Suppose that test scores independently follow an exponential distribution with PDF

$$f(x) = \frac{1}{\theta}e^{-x/\theta}, x > 0$$

Estimate θ using MLE.

Solution Start by calculating the likelihood function, which is the product of 33 likelihoods – one for each observation. Since some of the observations are repeated, unique likelihoods can be raised to the corresponding number of repeats.

$$\begin{aligned}
L(\theta) &= \prod_{i=1}^{33} f(x_i) \\
&= \left[\frac{1}{\theta}e^{-60/\theta}\right]^3 \left[\frac{1}{\theta}e^{-70/\theta}\right]^9 \left[\frac{1}{\theta}e^{-80/\theta}\right]^{12} \left[\frac{1}{\theta}e^{-90/\theta}\right]^7 \left[\frac{1}{\theta}e^{-100/\theta}\right]^2 \\
&= \left(\frac{1}{\theta}\right)^{3+9+12+7+2} e^{-[3(60)+9(70)+12(80)+7(90)+2(100)]/\theta} \\
&= \frac{1}{\theta^{33}} e^{-2,600/\theta}
\end{aligned}$$

We could take the derivative of $L(\theta)$ directly, but it would require the product rule. Instead, find the log-likelihood function.

$$l(\theta) = \ln[L(\theta)] = -33\ln[\theta] - \frac{2,600}{\theta}$$

Take the derivative of $l(\theta)$, set it equal to 0, and solve for θ to obtain the MLE estimate.

$$l'(\theta) = -\frac{33}{\theta} + \frac{2,600}{\theta^2} = 0$$

$$-33\theta + 2,600 = 0$$

$$\hat{\theta} = \frac{2,600}{33} = 78.788$$

[sq]

Example 2.2.2 Suppose that a health actuary is analyzing claim losses, and observes the following data:

Loss Amount	Frequency
1	57
1.5	43
2	21
2.5	5
3	2
3.5	1
4	1

She assumes every observation was independently drawn from a single-parameter Pareto distribution with PDF

$$f(x) = \frac{\alpha}{x^{\alpha+1}}, x > 1$$

Determine the MLE estimate of α .

Solution First, determine the likelihood function.

$$\begin{aligned}
L(\alpha) &= \alpha^{57} \cdot \frac{\alpha^{43}}{1.5^{43(\alpha+1)}} \cdot \frac{\alpha^{21}}{2^{21(\alpha+1)}} \cdot \frac{\alpha^5}{2.5^{5(\alpha+1)}} \cdot \frac{\alpha^2}{3^{2(\alpha+1)}} \cdot \frac{\alpha}{3.5^{\alpha+1}} \cdot \frac{\alpha}{4^{\alpha+1}} \\
&= \frac{\alpha^{130}}{1.5^{43(\alpha+1)} \cdot 2^{21(\alpha+1)} \cdot 2.5^{5(\alpha+1)} \cdot 3^{2(\alpha+1)} \cdot 3.5^{\alpha+1} \cdot 4^{\alpha+1}}
\end{aligned}$$

Next, determine the log-likelihood function.

$$\begin{aligned}
l(\alpha) &= \ln[L(\alpha)] \\
&= 130\ln[\alpha] - 43(\alpha + 1)\ln[1.5] - 21(\alpha + 1)\ln[2] - 5(\alpha + 1)\ln[2.5] \\
&\quad - 2(\alpha + 1)\ln[3] - (\alpha + 1)\ln[3.5] - (\alpha + 1)\ln[4]
\end{aligned}$$

Calculate its first derivative with respect to α , and set to equal 0. Upon solving for α , the estimate is

$$l'(\alpha) = \frac{130}{\alpha} - 43\ln[1.5] - 21\ln[2] - 5\ln[2.5] - 2\ln[3] - \ln[3.5] - \ln[4] = 0$$

$$\begin{aligned}
\hat{\alpha} &= \frac{130}{43\ln[1.5] + 21\ln[2] + 5\ln[2.5] + 2\ln[3] + \ln[3.5] + \ln[4]} \\
&= 3.139
\end{aligned}$$

[sq]

Coach's Remarks Occasionally, it helps to note whether a multiplicative constant can be factored from the likelihood function. This is because the multiplicative constant will not contribute to the derivative of the log-likelihood function. As such, you may drop the multiplicative constant from the start.

In this problem, note that

$$\begin{aligned}
L(\alpha) &= \frac{\alpha^{130}}{1.5^{43(\alpha+1)} \cdot 2^{21(\alpha+1)} \cdot 4^{\alpha+1}} \\
&= \frac{1}{1.5^{43} \cdot 2^{21} \cdot \dots \cdot 4} \left(\frac{\alpha^{130}}{1.5^{43\alpha} \cdot 2^{21\alpha} \cdot \dots \cdot 4^\alpha} \right) \\
&\quad \underset{c}{\sim} \quad \quad \quad \underset{L^*(\alpha)}{\sim} \\
l'(\alpha) &= \frac{d}{d\alpha} \ln[L(\alpha)] \\
&= \frac{d}{d\alpha} \ln[c \cdot L^*(\alpha)] \\
&= \frac{d}{d\alpha} (\ln[c] + \ln[L^*(\alpha)]) \\
&= 0 + \frac{d}{d\alpha} \ln[L^*(\alpha)]
\end{aligned}$$

Thus, $\frac{d}{d\alpha} \ln[L^*(\alpha)]$ will produce the same result as $l'(\alpha)$. Dropping the multiplicative constant could save some time and effort.

3.6 2.2.2 Grouped Data

We can expand this idea of likelihood to handle data that are grouped into intervals. Let's say an observation is recorded as being in the interval $(a, b]$ (instead of a specific value such as c). Then, the likelihood of this observation is $Pr(a < X \leq b) = F(b) - F(a)$, where $F(x)$ is the CDF of X .

By extension, for independently drawn "grouped" observations (i.e. the interval that each observation falls in is known), the likelihood function of generic parameter θ is

$$L(\theta) = \prod_{i=1}^n [F(b_i) - F(a_i)]$$

where the i th observation is recorded to be in the interval $(a_i, b_i]$.

Once $L(\theta)$ is found, apply the same strategy already discussed to estimate θ .

Example 2.2.3 The following table details how long it took 30 independent computers to finish running a particular script of code:

Number of Computers	Running Time (seconds)
3	$(0, 1]$
5	$(1, 3]$
8	$(3, 7]$
12	$(7, 15]$
2	$(15, 31]$

Suppose a Pareto distribution with the following CDF is used to model the running times:

$$F(x) = 1 - \left(\frac{1}{x+1} \right)^\alpha, x > 0$$

Estimate α using MLE.

Solution For generic constants a and b ,

$$\begin{aligned} F(b) - F(a) &= \left[1 - \left(\frac{1}{b+1} \right)^\alpha \right] - \left[1 - \left(\frac{1}{a+1} \right)^\alpha \right] \\ &= \left(\frac{1}{a+1} \right)^\alpha - \left(\frac{1}{b+1} \right)^\alpha \\ &= \frac{1}{(a+1)^\alpha} - \frac{1}{(b+1)^\alpha} \end{aligned}$$

Therefore, the likelihood function of α is

$$L(\alpha) = \left[1 - \frac{1}{2^\alpha} \right]^3 \left[\frac{1}{2^\alpha} - \frac{1}{4^\alpha} \right]^5 \left[\frac{1}{4^\alpha} - \frac{1}{8^\alpha} \right]^8 \left[\frac{1}{8^\alpha} - \frac{1}{16^\alpha} \right]^{12} \left[\frac{1}{16^\alpha} - \frac{1}{32^\alpha} \right]^2$$

From here, we could brute-force our way with arithmetic and calculus. However, we can reduce our efforts by substituting

$$\beta = \frac{1}{2^\alpha}$$

to express the likelihood function as a function of β .

$$\tilde{L}(\beta) = (1 - \beta)^3 (\beta - \beta^2)^5 (\beta^2 - \beta^3)^8 (\beta^3 - \beta^4)^{12} (\beta^4 - \beta^5)^2$$

Furthermore, we can express each of the five factors with only two terms, β and $1 - \beta$, raised to some power. For example, $(\beta^2 - \beta^3)^8 = \beta^{2(8)}(1 - \beta)^8$. As a result,

$$\begin{aligned} \tilde{L}(\beta) &= \beta^{5+2(8)+3(12)+4(2)} (1 - \beta)^{3+5+8+12+2} \\ &= \beta^{65} (1 - \beta)^{30} \end{aligned}$$

With this setup, the α that maximizes $L(\alpha)$ can be found by first calculating the β that maximizes $\tilde{L}(\beta)$, then plugging it into the equation $\beta = \frac{1}{2^\alpha}$. The reason why this works will be discussed at the end of the solution.

Proceed with the usual steps.

$$\tilde{l}(\beta) = \ln \left[\tilde{L}(\beta) \right] = 65 \ln[\beta] + 30 \ln[1 - \beta]$$

$$\begin{aligned} \tilde{l}'(\beta) &= \frac{65}{\beta} - \frac{30}{1-\beta} = 0 \\ 65(1-\beta) - 30\beta &= 0 \\ 65 - 95\beta &= 0 \\ \beta &= \frac{65}{95} \end{aligned}$$

Therefore,

$$\begin{aligned} \frac{1}{2^\alpha} &= \frac{65}{95} \\ 2^{-\alpha} &= \frac{65}{95} \\ -\alpha \ln[2] &= \ln \left[\frac{65}{95} \right] \\ \hat{\alpha} &= -\frac{\ln[65/95]}{\ln[2]} = 0.547 \\ &[sq] \end{aligned}$$

Coach's Remarks Applying the chain rule,

$$\begin{aligned} l'(\alpha) &= \tilde{l}'(\beta) \cdot \frac{d\beta}{d\alpha} = \tilde{l}'(\beta) \cdot \left(-\frac{\ln[2]}{2^\alpha} \right) \\ l'(\alpha) = 0 &\Rightarrow \tilde{l}'(\beta) \cdot \left(-\frac{\ln[2]}{2^\alpha} \right) = 0 \end{aligned}$$

For a Pareto distribution, $\alpha > 0$. Notice that no finite value of α can make $-\ln[2]/2^\alpha$ equal 0. Therefore, $-\ln[2]/2^\alpha$ cannot contribute to solving the equation that consequently simplifies to $\tilde{l}'(\beta) = 0$.

In general, when performing a $\eta = g(\theta)$ substitution, you should consider whether $\frac{d\eta}{d\theta}$ can result in 0 for some possible θ .

More Information The “take first derivative; set equal 0; solve” strategy may not always work. It takes for granted that $l(\theta)$ (or $L(\theta)$) has a global maximum at a critical point. Regardless, remember that the objective is to maximize $l(\theta)$ (or $L(\theta)$). If and when the default strategy fails, it may be possible to obtain an estimate by reasoning things through.

3.7 2.2.3 Other Comments

Similar to the other two methods of estimation, the MLE method can be used to estimate more than one parameter. In that case, the likelihood function will be multivariate – one variable for every parameter. Then, we take the first partial derivative of the likelihood (or log-likelihood) function with respect to each parameter, and set each partial derivative to equal 0. Solving for the parameter values that satisfy all the equations will produce the MLE estimates of those parameter.

However, not every setup is practical or possible to solve by hand. Advances in computer software have greatly lowered this hurdle, even if not all caveats are addressed.

While the topic of MLE goes much deeper, this is all we will cover for this course.

3.8 2.2 Overview

In several important ways, the *maximum likelihood estimation* (MLE) is considered a step-up from both the method of matching moments and percentile matching. It estimates parameters by maximizing what is called a “likelihood function”.

3.9 2.3.1 Properties Of An Estimator

3.9.1 Bias

The bias of an estimator is calculated as

$$\text{Bias}(\hat{\theta}) = E[\hat{\theta}] - \theta$$

This describes how close the estimator’s mean is to the true value of the parameter. $\hat{\theta}$ is an *unbiased estimator* when its bias is 0, i.e. when $E[\hat{\theta}] = \theta$. Otherwise, $\hat{\theta}$ is a biased estimator. Clearly, unbiasedness is a favorable quality, but remember this is only one aspect of an estimator.

Keep in mind that the value of θ is not known. This means calculating a bias could result in an expression instead of a number.

Ultimately, bias reveals whether an estimator is unbiased or not. If not, the resulting expression could tell us something about the behavior of the bias.

Example 2.3.1 Let $X_1, X_2, \dots, X_n$ be a random sample with common mean μ , and common variance σ^2 . The sample mean, $\bar{X}$, is used as an estimator for μ .

Determine if $\bar{X}$ is an unbiased estimator of μ .

Solution To calculate the bias of $\bar{X}$, first calculate its mean.

$$\begin{aligned} E[\bar{X}] &= E\left[\frac{\sum_{i=1}^n X_i}{n}\right] \\ &= \frac{E[\sum_{i=1}^n X_i]}{n} \\ &= \frac{E[X_1] + E[X_2] + \dots + E[X_n]}{n} \\ &= \frac{\mu + \mu + \dots + \mu}{n} \\ &= \frac{n\mu}{n} \\ &= \mu \end{aligned}$$

Since $E[\bar{X}] = \mu$, the bias of $\bar{X}$ is 0. Therefore, $\bar{X}$ is an unbiased estimator of μ .

[sq]

Coach’s Remarks There was no need to go through the steps if you had already memorized that $E[\bar{X}] = \mu$ for a random sample. Recall that we had mentioned this fact back in Section 1.2.1.

The next two examples may seem challenging, but rest assured you won’t be asked to perform calculations like these for this course. The purpose of the examples is to highlight a few key points, which we’ll make clear after Example 2.3.3.

Example 2.3.2 Let $X_1, X_2, \dots, X_n$ be a random sample with a known common mean μ , and common variance σ^2 . Define $\hat{\sigma}^2$ as an estimator for σ^2 , where

$$\hat{\sigma}^2 = \frac{\sum_{i=1}^n (X_i - \mu)^2}{n}$$

Determine if $\hat{\sigma}^2$ is an unbiased estimator of σ^2 .

Solution Before finding the expected value of $\hat{\sigma}^2$, let's rewrite it into a form easier to handle going forward.

$$\begin{aligned} \frac{\sum_{i=1}^n (X_i - \mu)^2}{n} &= \frac{\sum_{i=1}^n (X_i^2 - 2\mu X_i + \mu^2)}{n} \\ &= \frac{\sum_{i=1}^n X_i^2}{n} - \frac{2\mu(\sum_{i=1}^n X_i)}{n} + \frac{n\mu^2}{n} \\ &= \frac{\sum_{i=1}^n X_i^2}{n} - 2\mu\bar{X} + \mu^2 \end{aligned}$$

As a result,

$$\begin{aligned} E[\hat{\sigma}^2] &= E\left[\frac{\sum_{i=1}^n X_i^2}{n} - 2\mu\bar{X} + \mu^2\right] \\ &= \frac{E[\sum_{i=1}^n X_i^2]}{n} - 2\mu E[\bar{X}] + E[\mu^2] \\ &= \frac{n(\sigma^2 + \mu^2)}{n} - 2\mu(\mu) + \mu^2 \\ &= \sigma^2 \end{aligned}$$

where we used the following facts:

- $E[X^2] = Var[X] + (E[X])^2$
- For a random sample, $E[\bar{X}] = \mu$

Since $E[\hat{\sigma}^2] = \sigma^2$, the bias of $\hat{\sigma}^2$ is 0. Therefore, $\hat{\sigma}^2$ is an unbiased estimator of σ^2 .

[sq]

In the previous example, we unrealistically assumed that μ was known and proceeded to use it in our estimator. Let's see what happens if we use the sample mean, $\bar{X}$, as a substitute for μ .

Example 2.3.3 Let $X_1, X_2, \dots, X_n$ be a random sample with common mean μ , and common variance σ^2 . Define $\hat{\sigma}^2$ as an estimator for σ^2 , where

$$\hat{\sigma}^2 = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n}$$

Determine if $\hat{\sigma}^2$ is an unbiased estimator of σ^2 .

Solution Borrowing from our work in the previous example, note that

$$\begin{aligned} \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n} &= \frac{\sum_{i=1}^n X_i^2}{n} - 2\bar{X}\bar{X} + \bar{X}^2 \\ &= \frac{\sum_{i=1}^n X_i^2}{n} - \bar{X}^2 \end{aligned}$$

As a result,

$$\begin{aligned}
E[\hat{\sigma}^2] &= E\left[\frac{\sum_{i=1}^n X_i^2}{n} - \bar{X}^2\right] \\
&= \frac{E[\sum_{i=1}^n X_i^2]}{n} - E[\bar{X}^2] \\
&= \frac{E[\sum_{i=1}^n X_i^2]}{n} - \text{Var}[\bar{X}] - (E[\bar{X}])^2 \\
&= \frac{n(\sigma^2 + \mu^2)}{n} - \frac{\sigma^2}{n} - \mu^2 \\
&= \sigma^2 - \frac{\sigma^2}{n} \\
&= \sigma^2 \left(\frac{n-1}{n}\right)
\end{aligned}$$

where we used the following facts:

- $E[X^2] = \text{Var}[X] + (E[X])^2$
- For a random sample, $E[\bar{X}] = \mu$ and $\text{Var}[\bar{X}] = \frac{\sigma^2}{n}$

The bias of $\hat{\sigma}^2$ is

$$\sigma^2 \left(\frac{n-1}{n}\right) - \sigma^2 = -\frac{\sigma^2}{n} \neq 0$$

meaning $\hat{\sigma}^2$ is a biased estimator of σ^2 .

[sq]

In conclusion:

- The negative bias suggests that the average estimate would be lower than the true value of the parameter. But as n increases, notice that the bias converges to 0; this is what defines an *asymptotically unbiased* estimator.
- Notice that the biased estimator would be unbiased if multiplied by $\frac{n}{n-1}$.

$$E\left[\frac{n}{n-1}\hat{\sigma}^2\right] = \frac{n}{n-1}E[\hat{\sigma}^2] = \frac{n}{n-1} \cdot \sigma^2 \left(\frac{n-1}{n}\right) = \sigma^2$$

In addition, this unbiased estimator is

$$\frac{n}{n-1} \cdot \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n} = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n-1}$$

which is the S^2 formula introduced in Section 1.2.2, hence the name “unbiased sample variance”.

- The use of $\bar{X}$ in place of μ is where the bias emerges. This substitution imposes one mathematical constraint on the X_i ’s, which relates to how a denominator of $n-1$ would adjust the bias back to 0. This deals with a conceptual understanding of “degrees of freedom”.

3.9.2 Variance

By definition, variance measures the dispersion of a random variable from its mean. Hence, a larger variance of an estimator suggests that the estimate is prone to be a value further from the estimator’s mean. As a result, even an unbiased estimator may be unhelpful when the variance is large.

Similar to calculating the mean, keep in mind that the variance of an estimator could result in an expression instead of a number.

Example 2.3.4 Let $X_1, X_2, \dots, X_n$ be a random sample with common mean μ , and common variance σ^2 . Define $\hat{\mu}$ as an estimator for μ , where

$$\hat{\mu} = \frac{X_1 + X_n}{2}$$

Determine

1. if $\hat{\mu}$ is an unbiased estimator of μ .
2. the variance of $\hat{\mu}$.

Solution to (1)

$$\begin{aligned} E[\hat{\mu}] &= E\left[\frac{X_1 + X_n}{2}\right] \\ &= \frac{E[X_1 + X_n]}{2} \\ &= \frac{E[X_1] + E[X_n]}{2} \\ &= \frac{\mu + \mu}{2} \\ &= \frac{2\mu}{2} \\ &= \mu \end{aligned}$$

Since $E[\hat{\mu}] = \mu$, the bias of $\hat{\mu}$ is 0. Therefore, $\hat{\mu}$ is an unbiased estimator of μ .

[sq]

Solution to (2)

$$\begin{aligned} Var[\hat{\mu}] &= Var\left[\frac{X_1 + X_n}{2}\right] \\ &= \frac{Var[X_1 + X_n]}{2^2} \\ &= \frac{Var[X_1] + Var[X_n]}{4} \\ &= \frac{\sigma^2 + \sigma^2}{4} \\ &= \frac{2\sigma^2}{4} \\ &= \frac{\sigma^2}{2} \end{aligned}$$

[sq]

Coach's Remarks Recall that the variance of $\bar{X}$ is $\frac{\sigma^2}{n}$ for a random sample. With n exceeding 2 in this example, note that $\bar{X}$ would have a smaller variance than $\hat{\mu}$, even though both are unbiased estimators of μ . In that aspect, $\bar{X}$ would be preferable over $\hat{\mu}$.

This supports the intuition that the average of all n observations should do better at estimating the population mean, compared to the average of only the first and last observations (or any pair of observations, for that matter).

3.9.3 Mean Squared Error (MSE)

The *mean squared error* is also commonly used to determine the quality of an estimator. By definition,

$$MSE(\hat{\theta}) = E\left[(\hat{\theta} - \theta)^2\right]$$

Recall that the definition of a variance is similar to the formula above, where the parameter θ is replaced with the mean of the random variable ($E[\hat{\theta}]$ in this case). Thus, the MSE is a similar measure to the variance, where the “point of reference” is the true parameter value instead of the mean.

An alternate formula for the MSE is

$$MSE(\hat{\theta}) = Var[\hat{\theta}] + [Bias(\hat{\theta})]^2$$

The relation between MSE and variance is also seen in this formula. As mentioned, the distinction is whether the parameter θ or the mean $E[\hat{\theta}]$ is the “point of reference”. Therefore, when the estimator’s bias is 0 (i.e. $E[\hat{\theta}] = \theta$), the MSE will equal the variance.

Example 2.3.5 For independent random variables X_1, X_2, X_3 ,

- $E[X_i] = \frac{5-i}{4}\theta, i = 1, 2, 3$
- $Var[X_i] = (i+1)\theta^2, i = 1, 2, 3$
- An estimator for θ is

$$\hat{\theta} = \frac{4X_1 + 3X_2 + 2X_3}{9}$$

For $\theta > 0$, determine the range of θ values where the MSE of $\hat{\theta}$ is less than 100.

Solution Start by calculating the mean of $\hat{\theta}$.

$$\begin{aligned} E[\hat{\theta}] &= E\left[\frac{4X_1 + 3X_2 + 2X_3}{9}\right] \\ &= \frac{4E[X_1] + 3E[X_2] + 2E[X_3]}{9} \\ &= \frac{4(\theta) + 3(0.75\theta) + 2(0.5\theta)}{9} \\ &= \frac{29}{36}\theta \end{aligned}$$

Next, find the variance of $\hat{\theta}$.

$$\begin{aligned} Var[\hat{\theta}] &= Var\left[\frac{4X_1 + 3X_2 + 2X_3}{9}\right] \\ &= \frac{4^2 Var[X_1] + 3^2 Var[X_2] + 2^2 Var[X_3]}{9^2} \\ &= \frac{4^2(2\theta^2) + 3^2(3\theta^2) + 2^2(4\theta^2)}{9^2} \\ &= \frac{25}{27}\theta^2 \end{aligned}$$

Therefore,

$$\begin{aligned} MSE(\hat{\theta}) &= Var[\hat{\theta}] + [Bias(\hat{\theta})]^2 \\ &= \frac{25}{27}\theta^2 + \left[\frac{29}{36}\theta - \theta\right]^2 \\ &= \frac{25}{27}\theta^2 + \left[\frac{29}{36} - 1\right]^2\theta^2 \\ &= 0.9637\theta^2 \end{aligned}$$

$$\begin{aligned}
MSE(\hat{\theta}) &< 100 \\
\Rightarrow 0.9637\theta^2 &< 100 \\
\Rightarrow \theta^2 &< \frac{100}{0.9637} \\
\Rightarrow -\sqrt{\frac{100}{0.9637}} &< \theta < \sqrt{\frac{100}{0.9637}}
\end{aligned}$$

Since θ must be positive, the answer is

$$0 < \theta < 10.190 < \theta < 10.19$$

[sq]

3.9.4 Uniformly Minimum Variance Unbiased Estimator (UMVUE)

The *uniformly minimum variance unbiased estimator* is an unbiased estimator with the smallest variance among all unbiased estimators, regardless of the true parameter value. In a sense, the UMVUE is the best unbiased estimator.

To be clear, the UMVUE **does not** necessarily have the smallest variance of all estimators; it is only among the **unbiased** ones. Therefore, the variance of the UMVUE could possibly be larger than the variance of a biased estimator.

Finding or proving an estimator is the UMVUE is rather challenging, so it won't be covered in this course.

3.9.5 Consistency

Another way to examine an estimator is to see if it is *consistent*. By definition, $\hat{\theta}$ is said to be consistent if

$$\lim_{n \rightarrow \infty} Pr(|\hat{\theta} - \theta| > \varepsilon) = 0$$

for all $\varepsilon > 0$. Keep in mind that $\hat{\theta}$ is constructed using the sample size n .

Consistency assesses whether $\hat{\theta}$ converges (in probability) to θ when the sample size approaches ∞ .

In addition, if

- $\lim_{n \rightarrow \infty} Bias(\hat{\theta}) = 0$ (i.e. asymptotically unbiased), and
- $\lim_{n \rightarrow \infty} Var[\hat{\theta}] = 0$

then $\hat{\theta}$ is consistent. Otherwise, it is inconclusive.

Example 2.3.6 Let $X_1, X_2, \dots, X_n$ be a random sample with common mean β , and common variance $7\beta^2$. The sample mean, $\bar{X}$, is used as an estimator for β .

Determine whether $\bar{X}$ is consistent.

Solution We have already proven that $\bar{X}$ is an unbiased estimator of the mean (denoted as β here), which also implies that $\bar{X}$ is asymptotically unbiased. Mathematically,

$$\lim_{n \rightarrow \infty} Bias(\bar{X}) = \lim_{n \rightarrow \infty} 0 = 0$$

Also, recall that the variance of $\bar{X}$ is $\frac{7\beta^2}{n}$, since there is a random sample. Thus,

$$\lim_{n \rightarrow \infty} \text{Var} [\bar{X}] = \lim_{n \rightarrow \infty} \frac{7\beta^2}{n} = 0$$

As a result, $\bar{X}$ is **consistent**.

[sq]

Coach's Remarks Realize that $\bar{X}$ would still be consistent if the common mean had no relation with the common variance.

3.9.6 Efficiency

To compare two estimators, we may employ the idea of efficiency. For two generic estimators $\hat{\theta}_1$ and $\hat{\theta}_2$, the *efficiency* of $\hat{\theta}_1$ relative to $\hat{\theta}_2$ is calculated as

$$\text{Eff}(\hat{\theta}_1, \hat{\theta}_2) = \frac{\text{MSE}(\hat{\theta}_2)}{\text{MSE}(\hat{\theta}_1)}$$

The estimator in the denominator would be relatively more efficient if the relative efficiency is greater than 1, and vice versa. An easy way to remember this is to keep in mind that low MSEs are favorable, and thus, coincides with being more efficient.

$$\text{MSE}(\hat{\theta}_2) > \text{MSE}(\hat{\theta}_1) \Rightarrow \frac{\text{MSE}(\hat{\theta}_2)}{\text{MSE}(\hat{\theta}_1)} > 1$$

3.10 2.3 Overview

Now we take a step back to consider the theoretical implications of estimation. Our focus shifts away from the processes involving actual observed data that we saw in the previous subsections. Hence, it is important to distinguish between an “estimator” and an “estimate”.

An *estimator* is a rule (i.e. a function or a formula) that describes how to calculate an estimate. The main point is that an estimator studies the observations as **random variables**, i.e. before the actual values are seen.

The distinction between estimator and estimate can be further illustrated by taking a deeper look at several old examples. In each case, $X_1, X_2, \dots, X_n$ is a random sample (similar to our discussion in Section 1.2).

Revisiting Example 2.1.1 The method of matching moments was used to estimate θ for an exponential distribution.

The **estimator** for this setup is

$$\hat{\theta} = \frac{\sum_{i=1}^n X_i}{n}$$

Given the data of $n = 33$ students, the **estimate** of θ is 78.788.

Revisiting Example 2.1.2 The method of matching moments was used to estimate α and θ for a gamma distribution.

The **estimators** for this setup are

$$\hat{\theta} = \frac{(\sum_{i=1}^n X_i^2/n) - (\sum_{i=1}^n X_i/n)^2}{\sum_{i=1}^n X_i/n}$$

$$\hat{\alpha} = \frac{\sum_{i=1}^n X_i/n}{\hat{\theta}} = \frac{(\sum_{i=1}^n X_i/n)^2}{(\sum_{i=1}^n X_i^2/n) - (\sum_{i=1}^n X_i/n)^2}$$

Given the data of $n = 33$ students, the **estimates** of α and θ are 57.679 and 1.366, respectively.

Revisiting Example 2.2.2 The MLE was used to estimate α for a single-parameter Pareto distribution with lower bound of 1.

The **estimator** for this setup is

$$\hat{\alpha} = \frac{n}{\sum_{i=1}^n \ln[X_i]}$$

Given the data of $n = 130$ claims, the **estimate** of α is 3.139.

Relating to Section 1.2, estimators are statistics, i.e. functions of random variables. Hence, estimators are also random variables. The term “estimator” helps us remember that these statistics are specifically for estimating parameters.

This subsection covers the common terminologies and properties that relate to point estimators. We will use θ to denote a generic parameter of interest, and $\hat{\theta}$ to denote a generic estimator of θ . It may help to think of $\hat{\theta}$ as an arbitrary function of the random variables $X_1, X_2, \dots, X_n$, i.e. $\hat{\theta} = g(X_1, X_2, \dots, X_n)$.

How can we tell whether an estimator is good or bad? What criteria can we use to determine the quality of an estimator? We will cover the following properties of an estimator that can aid in evaluating its quality:

- Bias
- Variance
- Mean squared error
- Uniformly minimum variance unbiased estimator (UMVUE)
- Consistency
- Efficiency

4 3. Confidence Intervals

4.1 3.1.1 Basic About Confidence Interval

In this section, we will cover interval estimation for a parameter. The idea is to construct an interval known as a *confidence interval*. The parameter of interest lies within this interval with a certain level of confidence. Broadly speaking, we require the following to obtain a confidence interval:

- A methodology
- A suitable distribution
- A confidence level
- Data

In general, consider

$$Pr(L \leq \theta \leq U) = k$$

This describes a random interval (L, U) that encloses a generic parameter of interest θ . Notice that the random components above are L and U . In other words, θ is **not** random, which makes sense since it is a parameter. When data is collected (meaning values of random variables are observed) and plugged into L and U , the resulting numerical interval is a $100k\%$ confidence interval for θ . As such, k is known as the confidence level.

4.1.1 Pivotal Method

One of the most basic methods to construct a confidence interval is called the *pivotal method*. The method relies on a *pivotal quantity*, which is a function of the variables observed from a sample and the unknown parameters of the variables' distribution. This means a pivotal quantity is a random variable. Most importantly, the distribution of a pivotal quantity must not depend on the aforementioned parameters.

Coach's Remarks The definition for "pivotal quantity" is similar but not the same to the definition for "statistic" introduced in Section 1.2. Here is how they differ:

If a pivotal quantity consists of unknown parameter values, then the pivotal quantity is not a statistic.

If a statistic's sampling distribution depends on unknown parameters, then the statistic is not a pivotal quantity.

We will illustrate how this method works with the following trivial example.

Random variable X has CDF

$$F_X(x) = 1 - \left(\frac{\theta}{x + \theta} \right)^2, x > 0$$

A pivotal quantity is $Y = \frac{\theta}{\bar{X}}$, with CDF

$$F_Y(y) = \left(\frac{y}{y + 1} \right)^2, y > 0$$

noting that its distribution does not depend on θ .

Assuming X was observed to equal 3.5, use this pivotal quantity to construct a 95% confidence interval for the parameter θ .

Start with the following probability statement involving the pivotal quantity:

$$Pr(a \leq Y \leq b) = 0.95$$

Let a be the 2.5 th percentile of Y , and b the 97.5 th percentile of Y . Thus,

$$\left(\frac{a}{a+1} \right)^2 = 0.025 \Rightarrow a = 0.1878$$

$$\left(\frac{b}{b+1} \right)^2 = 0.975 \Rightarrow b = 78.4968$$

All that's left is to obtain the form $Pr(L \leq \theta \leq U) = 0.95$ from the initial probability statement.

$$\begin{aligned}
& Pr(0.1878 \leq Y \leq 78.4968) = 0.95 \\
\Rightarrow & Pr\left(0.1878 \leq \frac{\theta}{\bar{X}} \leq 78.4968\right) = 0.95 \\
\Rightarrow & Pr(0.1878\bar{X} \leq \theta \leq 78.4968\bar{X}) = 0.95
\end{aligned}$$

This means the random interval (L, U) is $(0.1878X, 78.4968X)$. Having observed X to be 3.5, the 95% confidence interval for θ is

$$(0.1878[3.5], 78.4968[3.5]) = (0.6573, 274.7389)$$

So instead of a point estimate for the unknown value of θ , we have produced an interval which – with reasonably high confidence – includes the value of θ . With this example, the key takeaways are:

- In general, L and U are functions of random variables and constants. The random variables are substituted by observed data to compute the confidence interval.
- The constants in L and U come from the desired confidence level and the pivotal quantity's distribution.

Furthermore, the *width* of a confidence interval is the length between its lower and upper bounds. For example, the width of $(0.6573, 274.7389)$ is $274.7389 - 0.6573 = 274.0816$.

More Information There is often confusion in interpreting a confidence interval. For the trivial example above, the temptation is to conclude that there is a 95% **probability** that θ is between 0.6573 and 274.7389; this is an **incorrect** interpretation of the interval. Remember that θ has a fixed (albeit unknown) value. The value simply is within the interval, or simply is not. There is no element of chance or randomness.

Instead, it is accurate to say that 95% of numerous, similarly constructed intervals would enclose θ . While $(0.6573, 274.7389)$ is one of these intervals, it is practically impossible to know whether it is among the 95% that would enclose θ , or the 5% that would not. This is what it means to be 95% confident.

This section will use the pivotal method to construct a confidence interval for the following parameters:

- Mean
- Difference in means
- Proportion
- Variance

4.2 3.1.2 Constructing Confidence Intervals

Start with a random sample $X_1, X_2, \dots, X_n$ with a common mean μ , and a common variance σ^2 . Our goal is to construct a confidence interval for μ .

If n is large enough, then by the Central Limit Theorem, we know the sample mean, $\bar{X}$, will have an approximate normal distribution, and in turn

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

would have a standard normal distribution. In fact, this Z is a pivotal quantity. Under this setup, the bounds of a 100% symmetric random interval (L, U) are:

$$L = \bar{X} + z_{(1-k)/2} \frac{\sigma}{\sqrt{n}}$$

$$U = \bar{X} + z_{(1+k)/2} \frac{\sigma}{\sqrt{n}}$$

where z_p is the 100 p th percentile of Z . In addition, realize that

$$z_{(1-k)/2} = -z_{(1+k)/2}$$

since the standard normal distribution is symmetric about 0. Therefore, a 100 k % confidence interval for μ is often given by the expression

$$\bar{x} \pm z_{(1+k)/2} \frac{\sigma}{\sqrt{n}}$$

where $\bar{x}$ is the sample mean calculated from the data.

Coach's Remarks To better understand the expression for this confidence interval, consider the following points:

- $z_{(1-k)/2}$ and $z_{(1+k)/2}$ produce a symmetric interval where Z is between them with 100 k % probability. This was alluded to in Example 1.2.5. The graph below illustrates the case when $k = 0.9$.

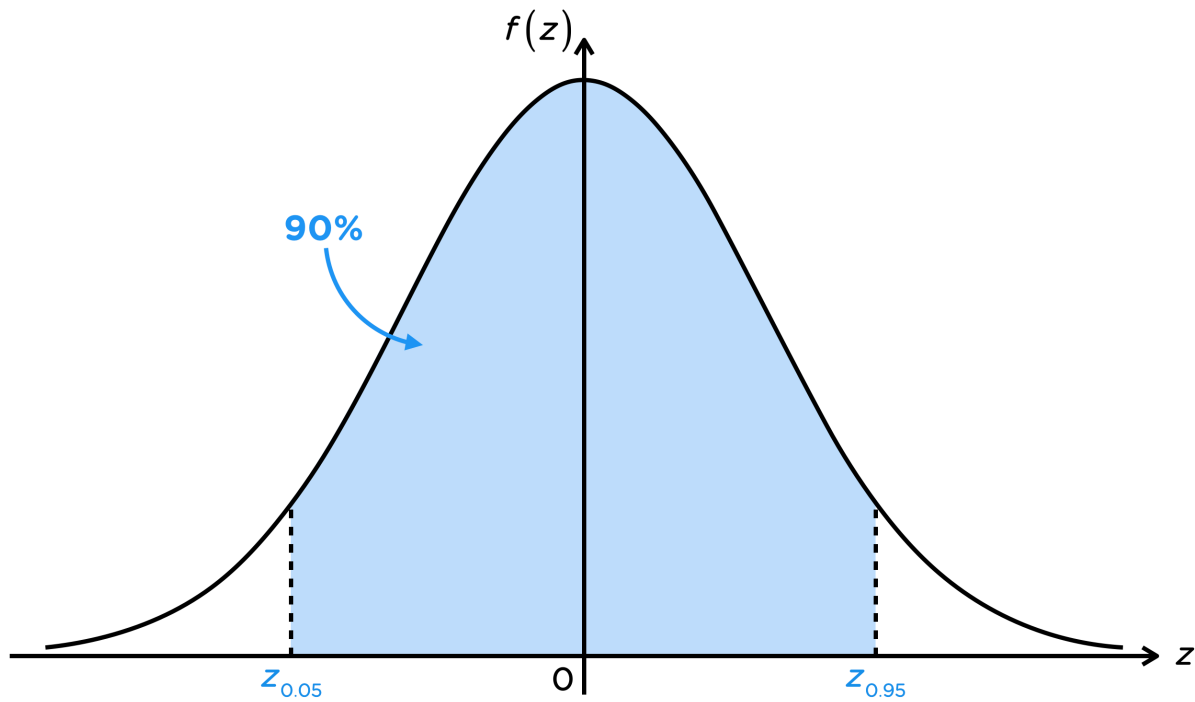


Figure 4: Line chart of the standard normal probability density function f of z versus z , displaying a symmetric bell curve centered at 0. The central 90 percent of the area is shaded between critical values z sub 0.05 on the left and z sub 0.95 on the right, marked by vertical dashed lines and labeled with a curved arrow as 90 percent.

- $\bar{X}$ and $\bar{x}$ are **different** . The first is a random variable, while the second is a calculated value (i.e. a constant) from observed data. They parallel how an estimator is different from an estimate.

However, what is the value of σ ? If it is unknown (as could very well be the case for a parameter), then the confidence interval expression cannot be used. Another issue is the situation where n is not large enough

for the Central Limit Theorem to apply – $\bar{X}$ may not have an approximate normal distribution. To account for all this, our discussion from Section 1.2.3 comes to the rescue.

To summarize, for a random sample of size n drawn from a normal distribution with common mean μ and common variance σ^2 ,

$$\frac{\bar{X} - \mu}{S/\sqrt{n}}$$

has a t -distribution with $n-1$ degrees of freedom. Similar to Z , this fraction is also a pivotal quantity. Given the similarities between the two setups, conclude that a $100k\%$ confidence interval for μ has the expression

$$\bar{x} \pm t_{(1+k)/2, n-1} \frac{s}{\sqrt{n}}$$

where

- $\bar{x}$ is the calculated sample mean,
- s is the calculated sample standard deviation, and
- $t_{p,\nu}$ is the $100p$ th percentile of a t random variable with ν degrees of freedom.

Coach's Remarks In this course, we will only focus on constructing confidence intervals for μ using the t -distribution. However, note that the sample must be drawn from a normal distribution for this to work. In contrast, the Central Limit Theorem from the first setup did not require a specific distribution.

Let's see how to construct these confidence intervals.

Example 3.1.1

After administering a standardized test, an exam committee obtained a random sample of 15 tests. Test scores are assumed to be normal distributed with mean μ . The sample average score and sample standard deviation of the 15 tests are 5 and 1.25, respectively.

Let $t_{p,\nu}$ be the $100p$ th percentile of a t random variable with ν degrees of freedom. The following table lists values of $t_{p,\nu}$ for specific combinations of p and ν :

	$p = 0.70$	$p = 0.80$	$p = 0.90$
$\nu = 14$	0.537	0.868	1.345
$\nu = 15$	0.536	0.866	1.341

Construct an 80% confidence interval for μ .

Solution The confidence interval expression for μ is

$$\bar{x} \pm t_{(1+k)/2, n-1} \frac{s}{\sqrt{n}}$$

We are given

$$n = 15, \bar{x} = 5, s = 1.25, k = 0.8$$

From the table, obtain $t_{(1+0.8)/2, 15-1} = t_{0.9, 14} = 1.345$.

Calculate the answer as

$$\left(5 - (1.345) \frac{1.25}{\sqrt{15}}, 5 + (1.345) \frac{1.25}{\sqrt{15}} \right) = (4.566, 5.434)$$

This interval means there is 80% confidence that the value of μ is between 4.566 and 5.434.

[sq]

Example 3.1.2

Suppose that the heights of residents in Town Q are normally distributed. A random sample of 9 individuals have height measurements $x_1, x_2, \dots, x_9$ that can be summarized as:

- $\bar{x} = 1.717$
- $\sum_{i=1}^9 (x_i - 1.717)^2 = 0.06$

Let $t_{p,\nu}$ be the 100 p th percentile of a t random variable with ν degrees of freedom. The following table lists values of $t_{p,\nu}$ for specific combinations of p and ν :

	$p = 0.925$	$p = 0.95$	$p = 0.975$
$\nu = 7$	1.617	1.895	2.365
$\nu = 8$	1.592	1.860	2.306

Construct a 90% confidence interval for the mean height of residents in Town Q.

Solution We are given

$$n = 9, \bar{x} = 1.717, k = 0.9$$

Calculate the sample standard deviation as

$$\begin{aligned} s &= \sqrt{s^2} \\ &= \sqrt{\frac{\sum_{i=1}^9 (x_i - \bar{x})^2}{n-1}} \\ &= \sqrt{\frac{0.06}{8}} \\ &= 0.0866 \end{aligned}$$

From the table, obtain $t_{(1+0.9)/2, 9-1} = t_{0.95, 8} = 1.86$.

Thus, the confidence interval is

$$\left(1.717 - (1.86) \frac{0.0866}{\sqrt{9}}, 1.717 + (1.86) \frac{0.0866}{\sqrt{9}} \right) = (1.663, 1.771)$$

[sq]

Example 3.1.3

For a random sample of size 6 drawn from a normal distribution, the following values were observed:

22.6 21.5 18.6 19.6 26.3 20.1

The upper bound of the $100k\%$ confidence interval for the mean is 23.7242.

Let $t_{p,\nu}$ be the $100p$ th percentile of a t random variable with ν degrees of freedom. The following table lists values of $t_{p,\nu}$ for specific combinations of p and ν :

	$p = 0.90$	$p = 0.95$	$p = 0.975$
$\nu = 5$	1.476	2.015	2.571
$\nu = 6$	1.440	1.943	2.447

Find the value of k .

Solution First, compute $\bar{x}$ and s with the observations.

$$\bar{x} = \frac{22.6 + 21.5 + 18.6 + 19.6 + 26.3 + 20.1}{6} = 21.45$$

$$s^2 = \frac{(22.6-21.45)^2 + (21.5-21.45)^2 + \dots + (20.1-21.45)^2}{6-1} = 7.643$$

$$s = \sqrt{7.643} = 2.7646$$

Applying the confidence interval expression,

$$\begin{aligned} \bar{x} + t_{(1+k)/2, n-1} \frac{s}{\sqrt{n}} &= 23.7242 \\ \Rightarrow 21.45 + t_{(1+k)/2, 5} \frac{2.7646}{\sqrt{6}} &= 23.7242 \\ \Rightarrow t_{(1+k)/2, 5} &= 2.015 \end{aligned}$$

From the table, conclude that

$$\frac{1+k}{2} = 0.95 \Rightarrow k = 0.9$$

$$[sq]$$

4.3 3.2.1 Confidence Interval For Difference Of Two Means

In research, there are many experiments where two (or more) groups are observed. For example, one group would be the control group, while the other group would undergo some kind of treatment. This way, the effect of a treatment can be measured. From a statistician's viewpoint, one way to measure the effect is to

1. consider the two groups of observations as being drawn from two random samples, and
2. examine if the means from the two random samples are significantly different.

“Equal means” is equivalent to “the difference in means equals 0”. Thus, when a confidence interval for the difference in two means does **not** include 0, we are confident that the two means are significantly different from each other.

The following diagram illustrates how the difference in two means can capture whether two random samples are drawn from distinct or identical distributions.

Consider the following setup:

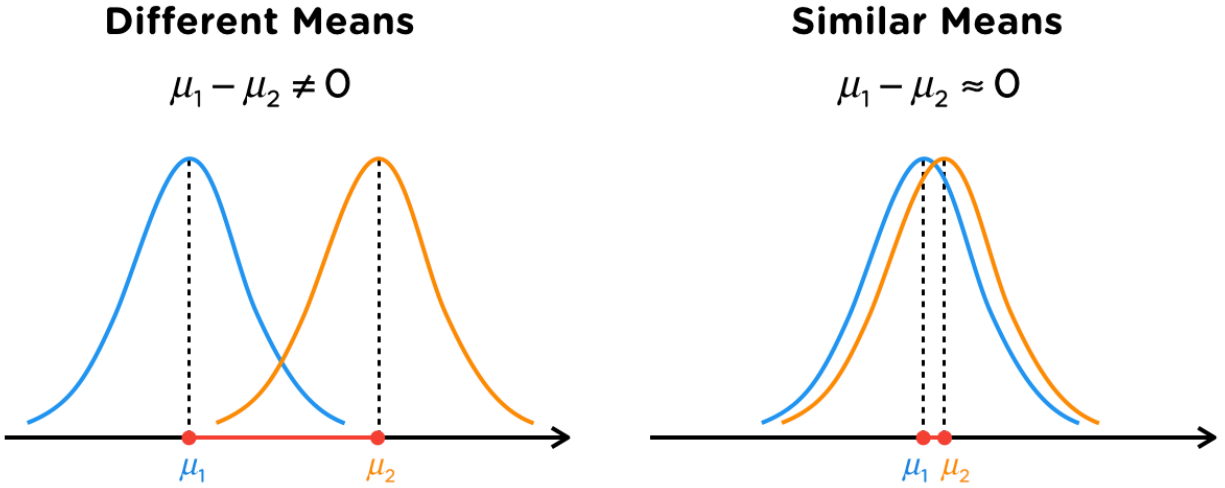


Figure 5: Two-panel normal distribution chart comparing two-sample means. Left panel titled Different Means shows separated blue and orange bell curves with equation $\mu_1 - \mu_2 \neq 0$ and dashed lines marking μ_1 and μ_2 , while right panel titled Similar Means shows overlapping curves with equation $\mu_1 - \mu_2 \approx 0$ and means marked closely together.

- There are two independent random samples.
- Random Sample 1 has size n_1 with common mean μ_1 and common variance σ_1^2 .
- Random Sample 2 has size n_2 with common mean μ_2 and common variance σ_2^2 .

Assuming the Central Limit Theorem applies to both samples, then their respective sample means, $\bar{X}_1$ and $\bar{X}_2$, as well as the difference in sample means, $\bar{X}_1 - \bar{X}_2$, all approximately follow a normal distribution, where

$$E [\bar{X}_1 - \bar{X}_2] = \mu_1 - \mu_2$$

$$Var [\bar{X}_1 - \bar{X}_2] = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}$$

This translates to a $100k\%$ confidence interval for $\mu_1 - \mu_2$ given by the expression

$$\bar{x}_1 - \bar{x}_2 \pm z_{(1+k)/2} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

But to use this, we are faced with similar issues presented in the previous subsection: we require the values of σ_1^2 and σ_2^2 , and the Central Limit Theorem must apply to both samples.

If the setup has two additional stipulations:

- Each random sample is drawn from a normal distribution
- Both normal distributions have the same variance (i.e. $\sigma_1^2 = \sigma_2^2$)

then we can make use of the t -distribution. The resulting $100k\%$ confidence interval expression for $\mu_1 - \mu_2$ would be

$$\bar{x}_1 - \bar{x}_2 \pm t_{(1+k)/2, n_1+n_2-2}(s_P) \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

The square of s_P (i.e. s_P^2) is the calculated *pooled sample variance*, which is a weighted average of the calculated sample variances.

$$s_P^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

More importantly, note that the **denominator** of the pooled sample variance matches the **degrees of freedom** of the t -distribution used here.

As a reminder,

- n_i is the sample size for sample i ,
- $\bar{x}_i$ is the calculated sample mean for sample i ,
- s_i^2 is the calculated sample variance for sample i , and
- $t_{p,\nu}$ is the 100 p th percentile of a t random variable with ν degrees of freedom.

Example 3.2.1

Scientists record the daily average temperatures for two independent cities. For each city, they randomly sample five days in June. Assume the two samples are drawn from two normal distributions having the same variance. The data is as follows:

City A: 78 77 75 71 74
City B: 73 78 86 83 85

Let $t_{p,\nu}$ be the 100 p th percentile of a t random variable with ν degrees of freedom. The following table lists values of $t_{p,\nu}$ for specific combinations of p and ν :

	$p = 0.95$	$p = 0.975$	$p = 0.99$
$\nu = 8$	1.860	2.306	2.896
$\nu = 9$	1.833	2.262	2.821

Construct a 90% confidence interval for City A's mean temperature minus City B's mean temperature.

Solution First, compute $\bar{x}_A$, $\bar{x}_B$, and s_P with the observations.

$$\bar{x}_A = \frac{78 + 77 + 75 + 71 + 74}{5} = 75$$

$$\bar{x}_B = \frac{73 + 78 + 86 + 83 + 85}{5} = 81$$

$$(n_A - 1)s_A^2 = (5 - 1) \cdot \frac{(78 - 75)^2 + (77 - 75)^2 + \dots + (74 - 75)^2}{5 - 1} = 30$$

$$(n_B - 1)s_B^2 = (5 - 1) \cdot \frac{(73 - 81)^2 + (78 - 81)^2 + \dots + (85 - 81)^2}{5 - 1} = 118$$

$$s_p = \sqrt{\frac{(n_A - 1)s_A^2 + (n_B - 1)s_B^2}{n_A + n_B - 2}} = \sqrt{\frac{30 + 118}{5 + 5 - 2}} = \sqrt{18.5}$$

From the table, obtain $t_{(1+0.9)/2, 5+5-2} = t_{0.95, 8} = 1.86$.

Applying the confidence interval expression,

$$\begin{aligned} & \bar{x}_A - \bar{x}_B \pm t_{(1+k)/2, n_A+n_B-2}(s_p) \sqrt{\frac{1}{n_A} + \frac{1}{n_B}} \\ \Rightarrow & 75 - 81 \pm 1.86 \sqrt{18.5} \sqrt{\frac{1}{5} + \frac{1}{5}} \\ \Rightarrow & -6 \pm 5.05975 \\ \Rightarrow & (-11.060, -0.940) \\ & [sq] \end{aligned}$$

4.3.1 Comments on the Example

Since the confidence interval does not include 0, one might conclude that the two cities have significantly different mean temperatures. However, here are some points for your own consideration:

- What does it mean for the cities to be independent? Must they be far apart to avoid having dependent weather behavior? How far is enough?
- Since both samples of five days were all taken in June, is it reasonable to claim that they are independent?
- Is it reasonable to assume a normal distribution for daily average temperatures?
- Note that $s_A^2 = 7.5$ and $s_B^2 = 29.5$. Is this evidence that the variances for both cities are actually not the same? Or do we simply have bad samples?

These points call to question the reliability of the confidence interval. While this course will not emphasize on scrutinizing the assumptions, it is instructive to not take them for granted.

4.4 3.2.2 Proportions

Numerous real-life events can be described using a Bernoulli distribution. Recall that it has a parameter p , representing the probability of a particular outcome (often referred to as a “success”), where only two outcomes are possible. Bernoulli applies to any situation that requires a yes/no or true/false response.

Assume a random sample $X_1, X_2, \dots, X_n$ is drawn from a Bernoulli distribution with common mean p and common variance $p(1 - p)$. The goal is to construct a confidence interval for p .

Once more, we can make use of the Central Limit Theorem. As a result, a $100k\%$ confidence interval expression for p is

$$\hat{p} \pm z_{(1+k)/2} \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

where $\hat{p}$ is the calculated sample mean of the observations. In addition, recall that a Bernoulli random variable equals 1 when a “success” occurs, or equals 0 otherwise. Therefore, $\hat{p}$ represents the proportion of “successes” from the data.

$$\hat{p} = \frac{\text{\# of "successes" from } n \text{ "trials"}}{n}$$

An equivalent setup is to have a single binomial observation. A binomial random variable implies the existence of a random sample drawn from a Bernoulli distribution. In this light, $\hat{p}$ equals the binomial outcome divided by n .

Unlike the previous two cases, there is no alternative if conditions for the Central Limit Theorem are not met. In other words, a large sample size is necessary for this confidence interval to be valid. The t -distribution cannot be used here, mainly because the random sample is drawn from a Bernoulli distribution instead of a normal distribution.

Example 3.2.2

To test whether a coin is biased, researchers made 100 independent tosses and recorded 30 heads.

Construct a 95% confidence interval for the probability of tossing a head with that coin.

Solution We are given

$$n = 100, \hat{p} = \frac{30}{100} = 0.3, k = 0.95$$

From a Z -table, note that $z_{(1+0.95)/2} = z_{0.975} = 1.96$.

This produces the interval

$$\left(0.3 - 1.96\sqrt{\frac{0.3 \cdot 0.7}{100}}, 0.3 + 1.96\sqrt{\frac{0.3 \cdot 0.7}{100}} \right) = (0.21, 0.39)$$

Based on the observed data, we have high confidence that the coin is biased, since the interval does not contain 0.5 – the value that represents a fair coin.

[sq]

More Information For the three cases we have discussed, the confidence interval expressions are similar, and not by coincidence. We may generalize as:

Consider an unbiased normally distributed estimator for parameter θ . A confidence interval for θ has the form:

$$\text{estimate} \pm (z\text{percentile})(\text{standard error})$$

where *standard error* is the standard deviation of the estimator. Using an estimator with a smaller standard error would produce a narrower and thus more informative confidence interval; using the UMVUE would be optimal.

However, the standard error may be a function of an unknown variance parameter. If the variance parameter can be replaced with an appropriate unbiased sample variance, the confidence interval form above uses a t percentile instead of a z percentile. In addition, the degrees of freedom for the t percentile will match the denominator of the unbiased sample variance.

4.5 3.2.3 Variances

Recall from Section 1.2.2 that for a random sample of size n drawn from a normal distribution with common mean μ and common variance σ^2 , the statistic

$$\frac{(n-1)S^2}{\sigma^2}$$

follows a χ^2 distribution with $n-1$ degrees of freedom. In fact, this statistic is also a pivotal quantity suitable for constructing a confidence interval for σ^2 .

Under this setup, a $100k\%$ confidence interval for σ^2 is

$$\left(\frac{(n-1)s^2}{\chi_{(1+k)/2, n-1}^2}, \frac{(n-1)s^2}{\chi_{(1-k)/2, n-1}^2} \right)$$

where

- s^2 is the calculated sample variance, and
- $\chi_{p, \nu}^2$ is the $100p$ th percentile of a χ^2 random variable with ν degrees of freedom.

Note that this confidence interval is already in its simplest expression, since a χ^2 distribution is **not** symmetric about 0 (in fact, it is right-skewed and only valid on the non-negative reals). Therefore, both percentiles in the interval must be determined individually.

Example 3.2.3 An astronomer studies how frequently a star is visible in a year. Assume that the number of appearances in a year is normally distributed. She randomly samples 8 years from a database and obtains the following data:

4 5 5 4 3 2 3 5

Let $\chi_{p, \nu}^2$ be the $100p$ th percentile of a chi-squared random variable with ν degrees of freedom. The following table lists values of $\chi_{p, \nu}^2$ for specific combinations of p and ν :

	$p = 0.025$	$p = 0.05$	$p = 0.95$	$p = 0.975$
$\nu = 6$	1.237	1.635	12.592	14.449
$\nu = 7$	1.690	2.167	14.067	16.013

Construct a 95% confidence interval for the variance of the number of appearances in a year.

Solution First calculate $(n-1)s^2$, which is the numerator of s^2 .

$$\bar{x} = \frac{4 + 5 + \dots + 5}{8} = 3.875$$

$$(n-1)s^2 = (4 - 3.875)^2 + (5 - 3.875)^2 + \dots + (5 - 3.875)^2 = 8.875$$

From the table, obtain

$$\chi_{(1+0.95)/2, 8-1}^2 = \chi_{0.975, 7}^2 = 16.013$$

$$\chi^2_{(1-0.95)/2, 8-1} = \chi^2_{0.025, 7} = 1.69$$

This produces the confidence interval

$$\left(\frac{8.875}{16.013}, \frac{8.875}{1.69} \right) = (0.554, 5.251)$$

[sq]

5 4. Hypothesis Testing

5.1 4.1.1 Properties Of Hypothesis Testing

A hypothesis test takes two opposing possibilities and checks which one is better supported by the available data. Specifically, the data is summarized by a single value which is judged to be either a plausible or implausible outcome using probability. A plausible value supports one possibility, while an implausible value supports the other. This logic is consistent with the intuition expressed in the introductory coin example.

To define the boundary that separates “plausible” from “implausible”, we need to be familiar with the terminology associated with hypothesis testing.

5.1.1 Null and Alternative Hypotheses

The two “opposing possibilities” mentioned are called the *null hypothesis* and the *alternative hypothesis*. They are often denoted as H_0 and H_1 , respectively. These hypotheses are usually mathematical statements about parameters of interest. For example, “a coin is fair” can be expressed as the hypothesis: a Bernoulli parameter $p = 0.5$.

The null hypothesis often takes a “status quo” position, meaning it is the statement assumed to be true by default. In turn, the alternative hypothesis is typically the statement that a researcher has interest in proving.

In conducting a hypothesis test, the calculations are performed assuming the null hypothesis is true. After weighing the evidence, the researcher decides to either:

- **Fail to reject** the null hypothesis, or
- **Reject** the null hypothesis in favor of the alternative hypothesis.

In other words, without sufficient evidence supporting H_1 , we keep assuming the default of H_0 . Otherwise, sufficient evidence favoring H_1 would suggest that H_0 ought to be rejected.

More Information Note that the first decision does **not** say “accept the null hypothesis”. Strongly affirming that something is true is arguably beyond the scope and ability of a hypothesis test. Thus, the phrase “fail to reject” is more accurate and preferred.

5.1.2 Test Statistic

The *test statistic* is a statistic as defined in Section 1.2, which is used to reject or not reject the null hypothesis. This is achieved by summarizing the sample observations while assuming the null hypothesis is true. Using its sampling distribution, we can determine whether the calculated test statistic from the data is considered “plausible” or “implausible”.

Coach’s Remarks The literature on statistics tends to use certain terms rather loosely. For example, both $\bar{X}$ and $\bar{x}$ are often simply referred to as “sample mean”, in spite of the inherent difference between the two, as previously discussed.

The term “test statistic” is no different. However, there is a lesser emphasis on a test statistic being a random variable in hypothesis testing. Therefore, we hereby use “test statistic” to only refer to the value calculated from the data.

If a test statistic is near either tail of the sampling distribution, then the data appears to be a rare occurrence (i.e. implausible). Conversely, a test statistic closer to the center of the sampling distribution suggests that the data appears to be a typical occurrence (i.e. plausible). Keep in mind that the sampling distribution is based on the null hypothesis being true.

When a test statistic is in either tail of the sampling distribution, perhaps it is not true that the data was a rare occurrence. Instead, the data may have come from a different distribution altogether. This implies that the null hypothesis is actually incorrect. In other words, an extreme test statistic would support the alternative hypothesis more than the null hypothesis.

To assist in learning the rest of the jargon, we will assume the following trivial setup throughout this subsection:

- There is one sample observation, X , with mean μ and variance σ^2 .
- The hypothesis test investigates the value of μ .
- The test statistic is x , the observed value of X .
- The sampling distribution is a normal distribution.

5.1.3 Rejection Region and Critical Value

The *rejection region* is the range of test statistic values that we consider “too extreme” and thus decide to reject H_0 in favor of H_1 . A *critical value* is a value that separates the rejection region from the rest of the possible test statistic values.

Coach’s Remarks Before continuing, it is important to distinguish between *two-tailed tests* and *one-tailed tests*.

- Two-tailed: both tails of the sampling distribution are included in the rejection region.
- One-tailed: only one tail of the sampling distribution is included in the rejection region.

We limit our discussion to two-tailed tests for now. The rejection region can be written as

$$[x \leq a] \cup [x \geq b]$$

meaning a test statistic x is “too extreme” if it is smaller than a **or** greater than b , in which case H_0 would be rejected. The critical values a and b are chosen such that both tails are symmetrical. Since X is normally distributed, the rejection region can also be written in terms of the standard normal distribution, such as

$$[z \leq -c] \cup [z \geq c] = |z| \geq c$$

where the test statistic is now $z = \frac{x-\mu}{\sigma}$. In this case, notice that we may avoid keeping track of two critical values, $-c$ and c , by taking the absolute value of z .

5.1.4 Significance Level

The critical value sets the boundary for how extreme the test statistic must be in order to reject the null hypothesis. A critical value is determined by setting a *significance level* denoted by α , where

$$Pr(|Z| \geq c \mid H_0 \text{ is true}) = \alpha$$

The significance level is the probability of rejecting H_0 , assuming it is true. Clearly, we would prefer not to reject H_0 if it is true, hence α is typically a small percent. The closer α is to 0, the less likely that H_0 will be rejected; a test statistic would need to be more extreme to provide evidence against H_0 .

The following graph illustrates the concepts. It shows a standard normal distribution assuming H_0 is true.

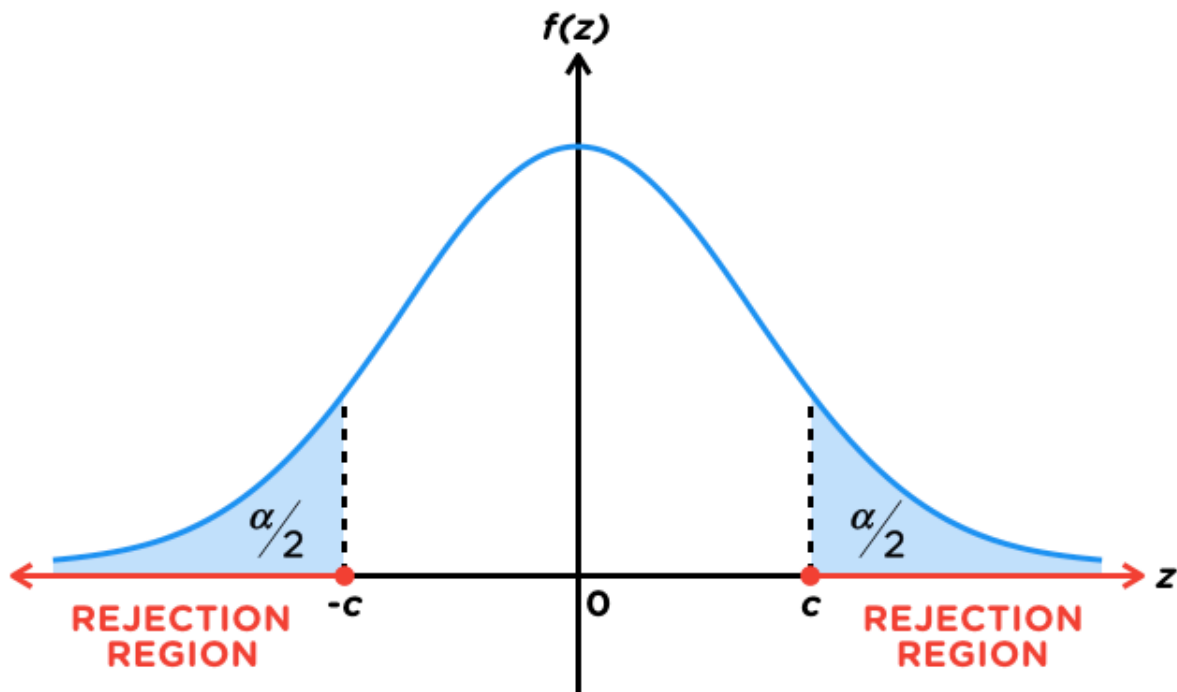


Figure 6: Line graph of a standard normal probability density function with horizontal axis z and vertical axis f of z . The symmetric bell curve has two shaded rejection regions: the left tail to the left of minus c with area α over 2 labeled REJECTION REGION, and the right tail to the right of c with area α over 2 labeled REJECTION REGION, with critical values minus c and c marked by vertical dashed lines and red dots on the horizontal axis.

From start to finish, a hypothesis test looks like this:

- Determine an appropriate significance level for the test. A common value is $\alpha = 0.05$.
- From the sampling distribution which assumes H_0 is true, determine the critical value that corresponds to the chosen α .
- Collect data, and use it to calculate the test statistic.
- Compare the test statistic to the critical value based on the rejection region. For a two-tailed test under the setup we have been assuming:
 - If $|z| \geq c$, then it is in the rejection region; reject H_0 .
 - If $|z| < c$, then it is not in the rejection region; do not reject H_0 .

Let's try performing a complete hypothesis test.

Example 4.1.1 It is believed that the mean age of licensed drivers in 2017 is 43.7. To test whether this is the case, one licensed driver's age is observed to be 60 in 2017. These ages are normally distributed with variance 80.

Test whether the mean age of licensed drivers in 2017 differs from 43.7 at the 5% significance level.

Solution Start by formally stating the null and alternative hypotheses. Letting μ represent the mean age of licensed drivers in 2017,

$$H_0 : \mu = 43.7$$

$$H_1 : \mu \neq 43.7$$

Next, determine the critical value. Since $\alpha = 0.05$,

$$Pr(|Z| \geq c) = 0.05$$

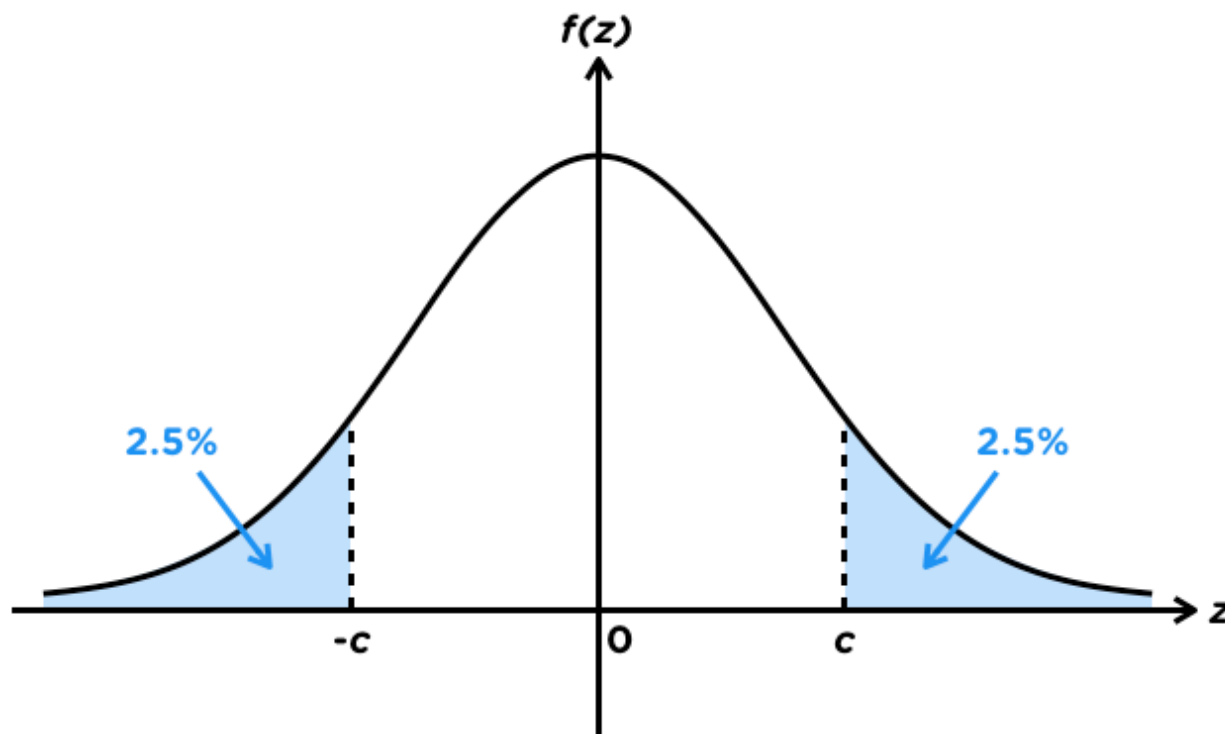


Figure 7: Line graph of the standard normal probability density function with vertical axis f of z and horizontal axis z . The symmetric bell curve centered at 0 has shaded tail regions beyond critical values at negative c and c , each labeled 2.5 percent.

The graph indicates that c is the $2.5 + 95 = 97.5$ th percentile of Z . From the Z -table, obtain $c = 1.96$.

Next, calculate the test statistic.

$$z = \frac{x - \mu}{\sigma} = \frac{60 - 43.7}{\sqrt{80}} = 1.82$$

Since $|1.82| < 1.96$, the test statistic does not fall in the rejection region, and thus we **do not reject the null hypothesis of $\mu = 43.7$ at the 5% significance level**. This means the observation of a 60-year-old licensed driver is plausible if the mean age is in fact 43.7.

[sq]

While no additional terminology is required to perform a hypothesis test, there are a few others worth mentioning.

5.1.5 p-Value

A *p-value* is the probability of observing the test statistic or a value more extreme, assuming H_0 is true. Thus, for a test statistic z , the *p-value* is

$$Pr(|Z| \geq |z| \mid H_0 \text{ is true})$$

This is similar to the definition of a significance level, with $|z|$ replacing the critical value c . Therefore, instead of comparing $|z|$ with c to make a decision, we may compare the *p-value* with the significance level α .

- If *p-value* $\leq \alpha$, then reject H_0 .
- If *p-value* $> \alpha$, then do not reject H_0 .

It should be evident that the two comparisons are equivalent. A larger $|z|$ results in a smaller *p-value*, and vice versa.

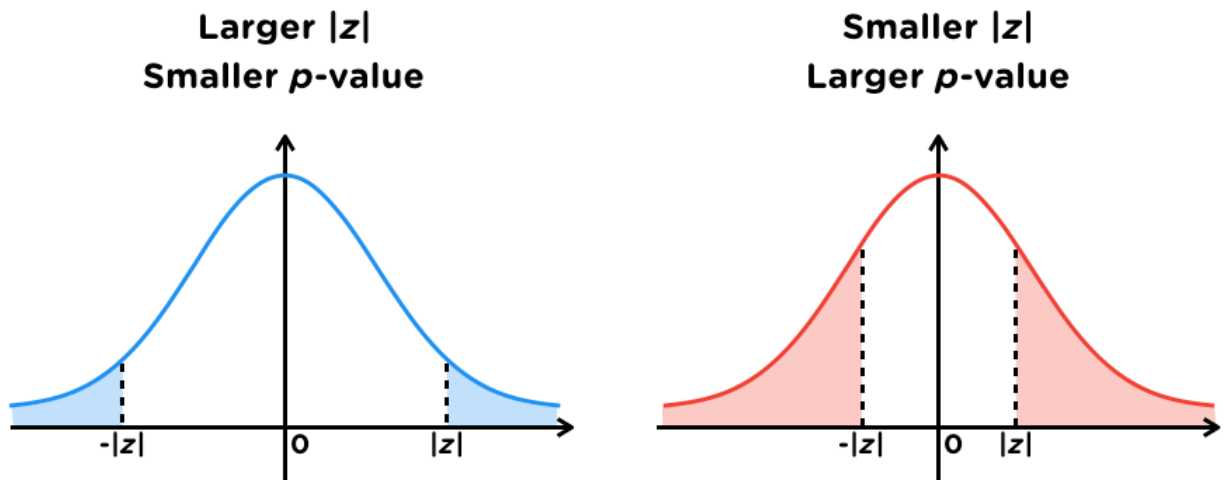


Figure 8: Two standard normal density curves side by side illustrating the relationship between the absolute value of z and *p* values in two-tailed tests. The left curve labeled Larger absolute value of z and Smaller *p* value shows small blue shaded tail areas beyond negative absolute value of z and absolute value of z , while the right curve labeled Smaller absolute value of z and Larger *p* value shows large red shaded tail areas beyond negative absolute value of z and absolute value of z .

What is the *p-value* for Example 4.1.1?

$$\begin{aligned} Pr(|Z| \geq |z|) &= 2 \cdot Pr(Z \geq |z|) \\ &= 2 \cdot Pr(Z \geq 1.82) \\ &= 2(1 - Pr(Z < 1.82)) \\ &= 2(1 - 0.9656) \\ &= 0.0688 \end{aligned}$$

As expected, we arrive at the same conclusion to not reject H_0 , since the *p-value* of 6.88% is greater than the 5% significance level.

5.1.6 Type I and Type II Errors

While we are unable to know for certain if a hypothesis is true, we try to make an informed decision with a hypothesis test. This does not mean hypothesis test decisions are always right, even when all the necessary

test assumptions are met. However, we can assume H_0 to be either true or false, and then consider the impact of making a wrong decision.

A *type I error* occurs when H_0 is rejected while it is true. The probability of making this error is the significance level α . In other words, we are willing to make a wrong decision $100\alpha\%$ of the time when H_0 is true, balanced by the possibility that H_0 is actually false.

A *type II error* occurs when H_0 fails to be rejected while it is false. The probability of making this error is denoted as β .

	H_0 is True	H_0 is False
Reject H_0	Type I Error	Correct Decision
Fail to Reject H_0	Correct Decision	Type II Error

In layman's terms, a type I error is a false positive, while a type II error is a false negative. Hypothesis test decisions are not error-free; they merely provide sensible judgment given the data.

5.1.7 Power of a Test

We would like to make the right decision of rejecting H_0 when it is false. The probability of making this right decision is called the *power of a test*, which is denoted by $1 - \beta$. This is because rejecting H_0 when it is false is the complement of a type II error.

Since the condition is that H_0 is false, a power can only be calculated when an explicit alternative to H_0 is provided.

What is the power of the test in Example 4.1.1 if in reality $\mu = 48$?

To do this, we need to find the critical values in the original unit of age.

$$\frac{a - 43.7}{\sqrt{80}} = -1.96 \Rightarrow a = 26.169$$

$$\frac{b - 43.7}{\sqrt{80}} = 1.96 \Rightarrow b = 61.231$$

In summary, the rejection region for the test is

$$[x \leq 26.169] \cup [x \geq 61.231]$$

Therefore, if $\mu = 48$, the power of the test is

$$\begin{aligned} Pr([X \leq 26.169] \cup [X \geq 61.231]) &= Pr\left(\left[Z \leq \frac{26.169-48}{\sqrt{80}}\right] \cup \left[Z \geq \frac{61.231-48}{\sqrt{80}}\right]\right) \\ &= Pr([Z \leq -2.44] \cup [Z \geq 1.48]) \\ &= Pr(Z \leq -2.44) + Pr(Z \geq 1.48) \\ &= [1 - Pr(Z < 2.44)] + 1 - Pr(Z < 1.48) \\ &= 1 - 0.9927 + 1 - 0.9306 \\ &= 0.0767 \end{aligned}$$

Intuitively, this low power makes sense. If $\mu = 48$, then H_0 should be rejected, since $\mu \neq 43.7$. However, there is not a major difference between $\mu = 43.7$ and $\mu = 48$. As a result, the test is unlikely to detect the distinction, and thus unlikely to correctly reject H_0 .

Other Information You may be curious what the rejection region is for the coin toss scenario. The null hypothesis is that $p = 0.5$ for a binomial sampling distribution with 10 “trials”. Note that

$$Pr(\leq 1\text{head}) + Pr(\geq 9\text{heads}) = 0.0215$$

while $Pr(\leq 2\text{heads}) + Pr(\geq 8\text{heads})$ would exceed 0.05. Therefore, at the 5% significance level, we reject the null hypothesis if 0, 1, 9, or 10 heads are observed from 10 coin tosses.

5.2 4.1.2 One Tailed Tests

As mentioned, a two-tailed test differs from a one-tailed test by the number of tails included to form the rejection region. This is driven by a researcher’s specific interest. In the case of a fair coin, **both** extremes of too few or too many heads would question whether the coin is fair. Thus, a two-tailed test is appropriate for that scenario. However, there are situations where **only one** extreme is of concern: too extreme to the left, or too extreme to the right, but not both.

For example, consider a factory claiming that the mean number of defective manufactured products per day is 4. The factory would typically be concerned if the mean is in fact much higher than 4, but not concerned if it is lower than 4. A one-tailed test is more suited in this scenario.

Before proceeding further, note that numerous different combinations of the null and alternative hypotheses fall under a one-tailed test. Despite these nuances, the core flow of a one-tailed test is the same. In general, to determine which type of test applies, we compare the two hypotheses to identify the appropriate rejection region.

5.2.1 Right-Tailed Tests

Let’s continue using our trivial setup. A possible set of hypotheses involving parameter μ and constant d is

$$H_0 : \mu \leq d \quad H_1 : \mu > d$$

An extreme x in the right tail of the normal distribution would support H_1 more than H_0 , but an extreme x in the left tail would not. This means the rejection region is in the right tail, given by

$$x \geq b \Rightarrow z \geq c$$

As expected, the critical value is determined by the significance level, where

$$Pr(Z \geq c \mid H_0 \text{ is true}) = \alpha$$

As seen in the graph below, α is **not** split into two regions. Therefore, for the same α , the critical value here will be different from the critical value in the two-tailed case.

Although H_0 does not specify an exact value for μ , use the exact point that separates the two hypotheses, i.e. use $\mu = d$.

In summary, here are the changes for this case:

- The rejection region only involves the right tail of the sampling distribution.
- The significance level is $Pr(Z \geq c \mid H_0 \text{ is true})$.
- The decision logic is:

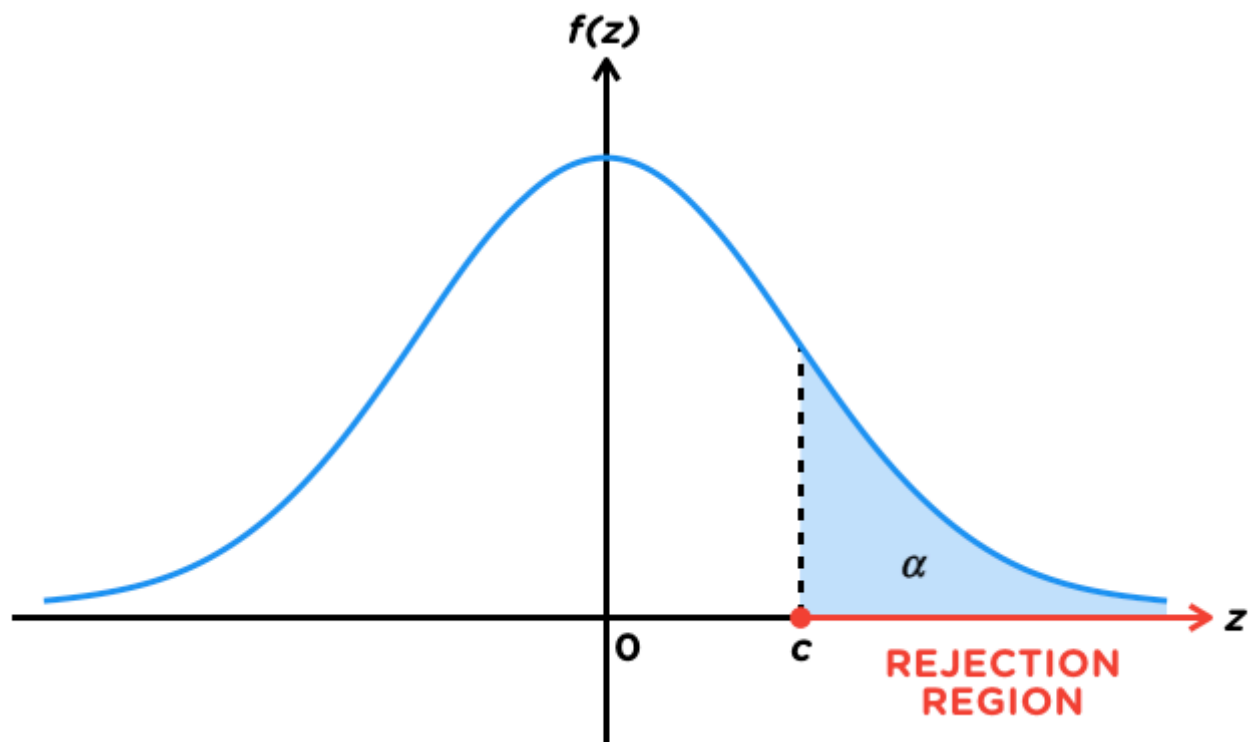


Figure 9: Line graph of a standard normal probability density function f of z versus z , showing a symmetric bell curve centered at 0. The right tail beyond critical value c is shaded and labeled α , with REJECTION REGION written in red below the horizontal axis indicating the rejection region for this one-tailed test.

- If $z \geq c$ for $\mu = d$, then $z \geq c$ will hold for all other possible μ values in H_0 . Therefore, reject H_0 .
- If $z < c$ for $\mu = d$, then the test statistic is not extreme enough for some subset of possible μ values in H_0 ; there's not enough evidence to deem H_0 unreasonable. Therefore, do not reject H_0 .
- The p -value is $Pr(Z \geq z \mid H_0 \text{ is true})$.

Let's try a few examples. The first is a variation on Example 4.1.1.

Example 4.1.2 It is believed that the mean age of licensed drivers in 2017 is 43.7. A researcher thinks that the mean age is actually higher. One licensed driver's age is observed to be 60 in 2017. These ages are normally distributed with variance 80.

Test whether the researcher is justified at the 5% significance level.

Solution Letting μ represent the mean age of licensed drivers in 2017, the null and alternative hypotheses are

$$H_0 : \mu = 43.7$$

$$H_1 : \mu > 43.7$$

Although H_0 differs from the form we introduced, this is still a right-tailed test. This is because H_1 would be supported by observing an extreme **older** age, thus justifying the researcher's claim. On the other hand, observing an extreme **younger** age would not justify the researcher's claim.

Next, determine the critical value. Since $\alpha = 0.05$,

$$Pr(Z \geq c) = 0.05$$

The graph indicates that c is the 95 th percentile of Z . From the Z -table, $c = 1.645$.

Next, calculate the test statistic.

$$z = \frac{x - \mu}{\sigma} = \frac{60 - 43.7}{\sqrt{80}} = 1.82$$

Since $1.82 > 1.645$, the test statistic falls in the rejection region, and thus we **reject the null hypothesis of $\mu = 43.7$ in favor of $\mu > 43.7$ at the 5% significance level**.

[sq]

Coach's Remarks Recall that for Example 4.1.1, H_0 was not rejected at the same 5% significance level. This is because when including both tails, the driver's age of 60 is not among the "top 5% of extreme values".

However, when including only the right tail, 60 is among the "top 5% of extreme values", thus H_0 is rejected here.

Example 4.1.3 X is normally distributed with mean μ and variance 22. The following set of hypotheses are considered:

- $H_0 : \mu = 5$
- $H_1 : \mu = 10$

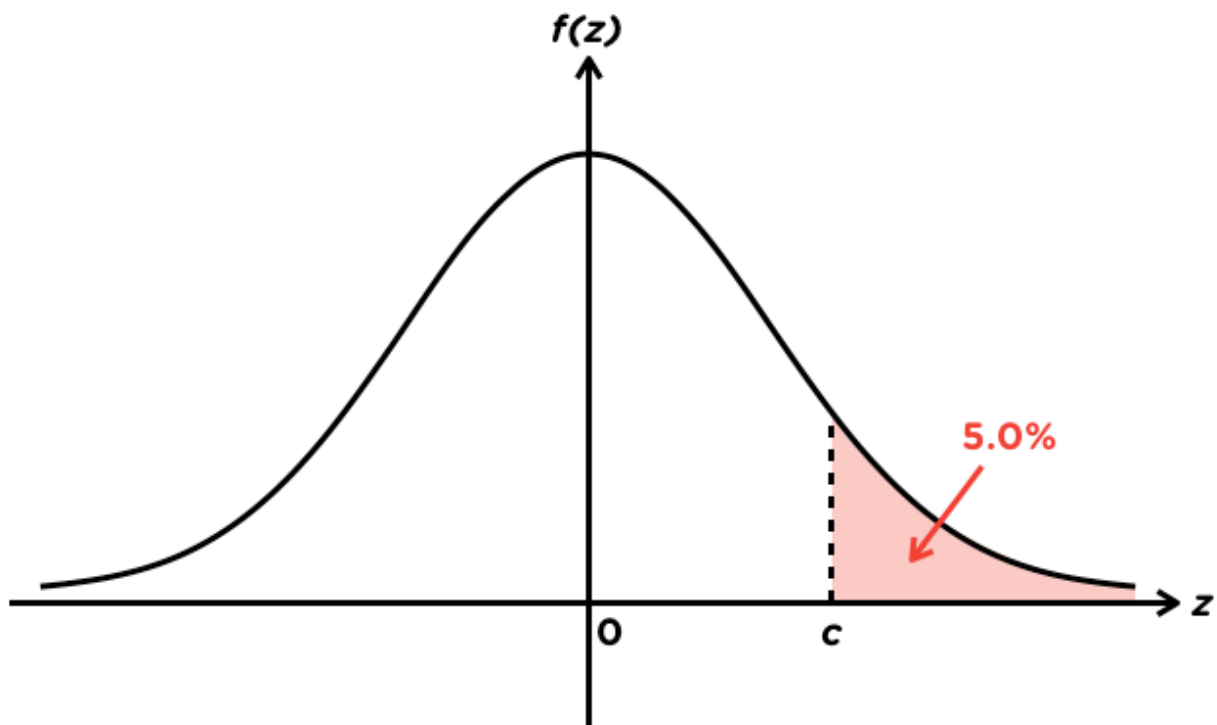


Figure 10: Line graph of a standard normal probability density function with horizontal axis z and vertical axis f of z . The symmetric bell curve is centered at 0 with the right tail region beyond critical value c shaded and labeled 5.0 percent, illustrating the rejection region for an upper one-tailed hypothesis test.

With a single observation and at the 5% significance level, calculate the power of this test.

Solution This time, both H_0 and H_1 differ from the form we introduced. Nevertheless, this is still a right-tailed test. This is because the μ hypothesized by H_1 is larger than the one by H_0 .

First, determine the critical value in the unit of X . For $\alpha = 0.05$, we previously saw that $c = 1.645$. Therefore,

$$\frac{b - 5}{\sqrt{22}} = 1.645 \Rightarrow b = 12.716$$

Hence, the rejection region is $x \geq 12.716$.

It is implied that the power is calculated based on H_1 . This produces the answer of

$$\begin{aligned} Pr(X \geq 12.716) &= Pr\left(Z \geq \frac{12.716 - 10}{\sqrt{22}}\right) \\ &= Pr(Z \geq 0.58) \\ &= 1 - Pr(Z < 0.58) \\ &= 1 - 0.719 \\ &= 0.281 \end{aligned}$$

[sq]

5.2.2 Left-Tailed Tests

Now consider the hypotheses

$$H_0 : \mu \geq d \quad H_1 : \mu < d$$

An extreme x in the left tail of the normal distribution would support H_1 more than H_0 , but an extreme x in the right tail would not. This means the rejection region is in the left tail, given by

$$x \leq a \Rightarrow z \leq -c$$

where the significance level is

$$Pr(Z \leq -c \mid H_0 \text{ is true}) = \alpha$$

Similar to the previous case, use $\mu = d$ in the calculations when assuming H_0 as true.

In summary, here are the changes for this case:

- The rejection region only involves the left tail of the sampling distribution.
- The significance level is $Pr(Z \leq -c \mid H_0 \text{ is true})$.
- The decision logic is:
 - If $z \leq -c$ for $\mu = d$, then $z \leq -c$ will hold for all other possible μ values in H_0 . Therefore, reject H_0 .
 - If $z > -c$ for $\mu = d$, then the test statistic is not extreme enough for some subset of possible μ values in H_0 ; there's not enough evidence to deem H_0 unreasonable. Therefore, do not reject H_0 .
- The p -value is $Pr(Z \leq z \mid H_0 \text{ is true})$.

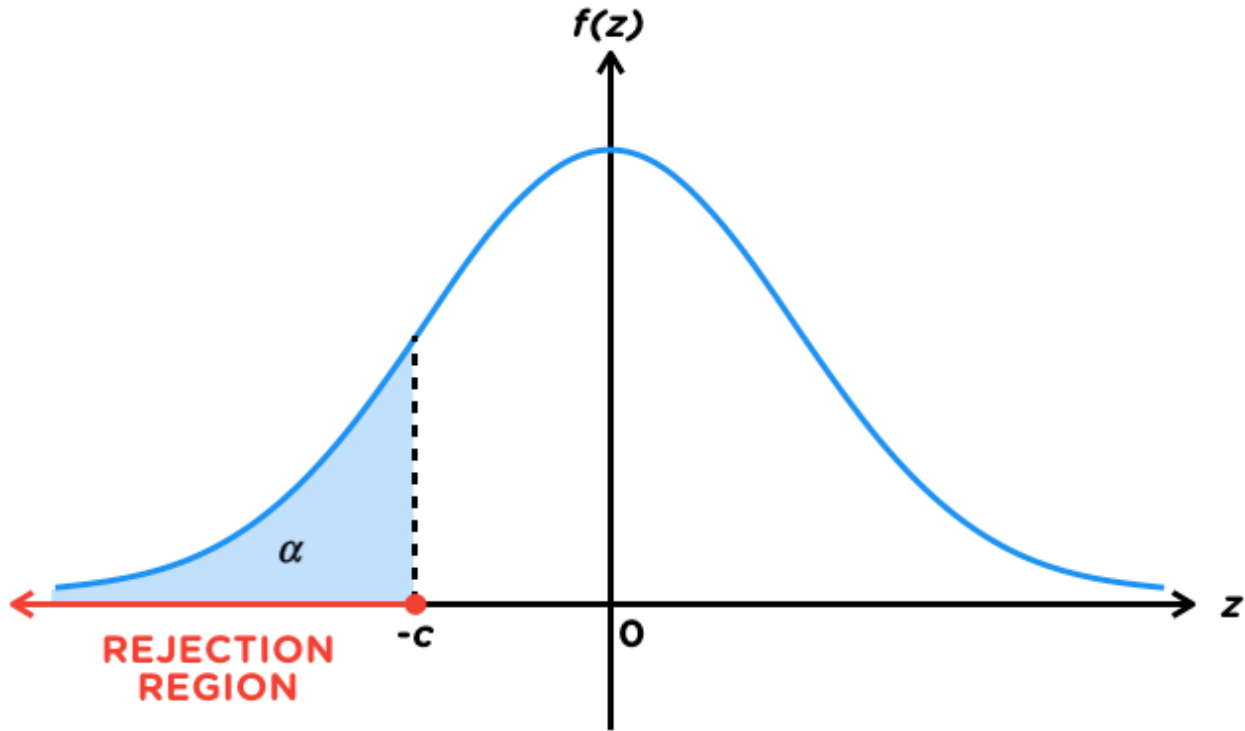


Figure 11: Line graph of a standard normal probability density function with z on the horizontal axis and f of z on the vertical axis. The left tail region less than negative c is shaded and labeled α , with an arrow indicating this is the rejection region.

Example 4.1.4 The mean total cholesterol for overweight individuals is said to be at least 250mg/dL. A drug chemist thinks the mean total cholesterol is actually lower. An overweight individual is observed to have a total cholesterol of 208mg/dL. Assume total cholesterol for these individuals is distributed normally with variance v .

At the 3% significance level, determine the largest value of v that would result in supporting the chemist.

Solution Letting μ represent the mean total cholesterol of overweight individuals, the null and alternative hypotheses are

$$H_0 : \mu \geq 250$$

$$H_1 : \mu < 250$$

Since $\alpha = 0.03$,

$$Pr(Z \leq -c) = 0.03$$

The graph indicates that c (as opposed to $-c$) would be the 97th percentile of Z . From the Z -table, $c = 1.88$. In turn, the critical value $-c = -1.88$.

To support the chemist (i.e. reject H_0), the test statistic must be less than or equal to the critical value. Remembering that $\mu = 250$ for the test statistic,

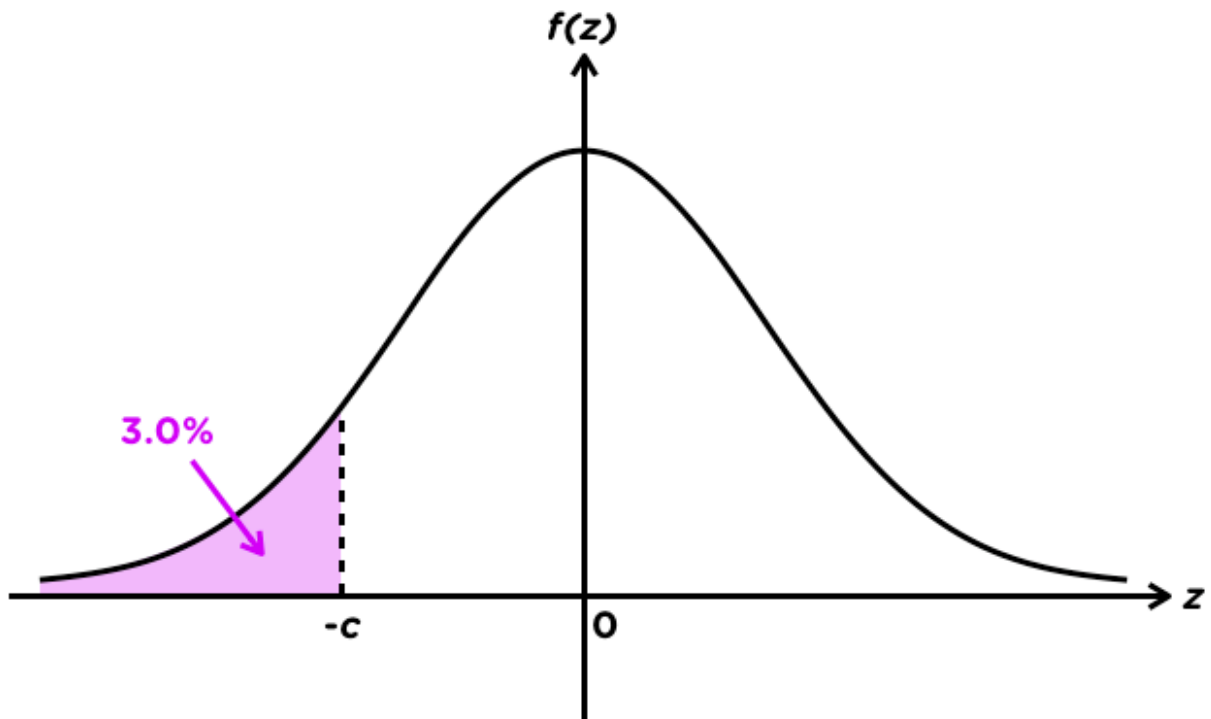


Figure 12: Line graph of a standard normal probability density function with horizontal axis z and vertical axis f of z . The shaded left tail area to the left of negative c is labeled 3.0 percent, representing the probability that z is less than negative c .

$$\begin{aligned}
\frac{208-250}{\sqrt{v}} &\leq -1.88 \\
\Rightarrow -\frac{42}{\sqrt{v}} &\leq -1.88 \\
\Rightarrow \frac{42}{\sqrt{v}} &\geq 1.88 \\
\Rightarrow \sqrt{v} &\leq \frac{42}{1.88} \\
\Rightarrow v &\leq (22.3404)^2
\end{aligned}$$

The largest value of v that would result in rejecting H_0 is approximately 499 .

[sq]

The following table summarizes the three variants of a hypothesis test, including the critical value formulas in terms of z_p , the $100p$ th percentile of Z . Remember that z without a subscript is the test statistic, and α is the significance level.

	Left-Tailed	Two-Tailed	Right-Tailed
Reject H_0 if	$z \leq -c$	$\ z\ \geq c$	$z \geq c$
Critical Value	$-z_{1-\alpha}$	$z_{1-\frac{\alpha}{2}}$	$z_{1-\alpha}$

More Information For left-tailed tests, we denote the critical value as $-c$, **not** c . Other resources may use different notation. For example, they may use only one symbol to represent the critical value, regardless of which variant is being discussed.

Ultimately, it is simply a different way to express the same thing. If you do consult other resources, pay close attention to how things are defined.

5.3 4.1 Overview

Consider testing whether a coin is fair. We saw one method from Example 3.2.2 using a confidence interval for proportions. However, that relies on the Central Limit Theorem. Could this be tested using only a few coin tosses?

If a coin was tossed 10 times, there are a few clear cases. If 5 heads were observed, it seems reasonable to claim that the coin is fair; if 0 or 10 heads were observed, it seems reasonable to claim that the coin is biased. What about the other possibilities, such as 3 heads?

Hypothesis tests aim to answer questions similar to this one by applying probability concepts. In particular, these tests provide statistical inference for a parameter (similar to point estimation and interval estimation) or for model evaluation.

5.4 4.2.1 Hypothesis Test For Means

For a random sample of size n drawn from a normal distribution with common mean μ and common variance σ^2 ,

$$\frac{\bar{X} - \mu}{S/\sqrt{n}}$$

has a t -distribution with $n - 1$ degrees of freedom. This is the same setup for constructing a confidence interval for means in Section 3.1.2.

As a result, a hypothesis test for μ has a test statistic calculated as

$$\frac{\bar{x} - \mu}{s/\sqrt{n}}$$

which comes from a t sampling distribution with $n - 1$ degrees of freedom.

Since both standard normal and t distributions are symmetric about 0, apply the same method in determining the critical value. If $t_{p,\nu}$ is the 100 p th percentile of a t random variable with ν degrees of freedom,

	Left-Tailed	Two-Tailed	Right-Tailed
Critical Value	$-t_{1-\alpha,n-1}$	$t_{1-\frac{\alpha}{2},n-1}$	$t_{1-\alpha,n-1}$

Example 4.2.1 Scientists research the claim that the mean height of North Linden residents is 1.67. Believing it is in fact higher, they conduct a hypothesis test by randomly sampling ten residents and measuring their heights. The sample average and sample variance of the ten heights are 1.86 and 0.1, respectively.

Let $t_{p,\nu}$ be the 100 p th percentile of a t random variable with ν degrees of freedom. The following table lists values of $t_{p,\nu}$ for specific combinations of p and ν :

	$p = 0.95$	$p = 0.975$	$p = 0.99$
$\nu = 9$	1.833	2.262	2.821
$\nu = 10$	1.812	2.228	2.764

Assuming heights are normally distributed, determine the test result.

Solution For μ representing the mean height of North Linden residents, the hypotheses are

$$H_0 : \mu = 1.67 \quad H_1 : \mu > 1.67$$

We are given

$$n = 10, \bar{x} = 1.86, s^2 = 0.1$$

The test statistic is

$$\frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{1.86 - 1.67}{\sqrt{0.1}/\sqrt{10}} = 1.9$$

Since we are not given a specific significance level, use the percentiles given in the table as the critical values. Since this test involves $10 - 1 = 9$ degrees of freedom, we only consider the row where $\nu = 9$, and note that

$$1.833 < 1.9 < 2.262$$

The test statistic is greater than a critical value of 1.833, but less than a critical value of 2.262. Determine the significance levels associated with these two values. Recall this is a right-tailed test.

$$\begin{aligned} 1.833 = t_{0.95,9} = t_{1-\alpha,9} &\Rightarrow \alpha = 1 - 0.95 = 0.05 \\ 2.262 = t_{0.975,9} = t_{1-\alpha,9} &\Rightarrow \alpha = 1 - 0.975 = 0.025 \end{aligned}$$

In conclusion, **reject H_0 at the 5% significance level, but not at the 2.5% level.**

[sq]

Example 4.2.2 An automated candy dispenser is used in a factory line. The weight of dispensed candy per bag is normally distributed with an assumed mean of 46oz. Through a hypothesis test, a technician performs a routine check to see if the mean dispensed weight differs from 46oz. The following are the weights of six randomly sampled bags:

42 46 43 43 44 46

Let $t_{p,\nu}$ be the 100 p th percentile of a t random variable with ν degrees of freedom. The following table lists values of $t_{p,\nu}$ for specific combinations of p and ν :

	$p = 0.975$	$p = 0.98$	$p = 0.985$
$\nu = 5$	2.571	2.757	3.003
$\nu = 6$	2.447	2.612	2.829

Determine the test result.

Solution For μ representing the mean dispensed weight, the hypotheses are

$$H_0 : \mu = 46 H_1 : \mu \neq 46$$

First, compute $\bar{x}$ and s from the observations.

$$\bar{x} = \frac{42 + 46 + 43 + 43 + 44 + 46}{6} = 44$$

$$s^2 = \frac{(42 - 44)^2 + (46 - 44)^2 + \dots + (46 - 44)^2}{6 - 1} = 2.8$$

$$s = \sqrt{2.8}$$

The test statistic is

$$\frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{44 - 46}{\sqrt{2.8}/\sqrt{6}} = -2.928$$

As this is a two-tailed test, remember to take the absolute value of the test statistic. Since this test involves $6 - 1 = 5$ degrees of freedom, we only consider the row where $\nu = 5$, and note that

$$2.757 < |-2.928| < 3.003$$

Determine the significance levels associated with 2.757 and 3.003.

$$\begin{aligned} 2.757 &= t_{0.98,5} = t_{1-\frac{\alpha}{2},5} \Rightarrow \alpha = 2(1 - 0.98) = 0.04 \\ 3.003 &= t_{0.985,5} = t_{1-\frac{\alpha}{2},5} \Rightarrow \alpha = 2(1 - 0.985) = 0.03 \end{aligned}$$

In conclusion, **reject H_0 at the 4% significance level, but not at the 3% level.**

[sq]

Example 4.2.3 A random sample of size 4 is taken from a normal distribution to test whether the mean is 50. The sample standard deviation of the 4 observations is 5.4224.

Let $t_{p,\nu}$ be the 100 p th percentile of a t random variable with ν degrees of freedom. The following table lists values of $t_{p,\nu}$ for specific combinations of p and ν :

	$p = 0.97$	$p = 0.975$	$p = 0.98$
$\nu = 2$	3.896	4.303	4.849
$\nu = 3$	2.951	3.182	3.482

If the null hypothesis is rejected when the sample mean of the 4 observations is either at most 42 or at least 58, determine significance level of the test.

Solution For μ representing the mean of the normal distribution, the hypotheses are

$$H_0 : \mu = 50 \quad H_1 : \mu \neq 50$$

This is a two-tailed test, since the rejection region includes both tails. The right side of the rejection region is

$$\bar{x} \geq 58 \Rightarrow \frac{\bar{x} - 50}{5.4224/\sqrt{4}} \geq t_{1-\frac{\alpha}{2},3}$$

In other words, 58 is a critical value in the unit of $\bar{X}$. Therefore,

$$t_{1-\frac{\alpha}{2},3} = \frac{58 - 50}{5.4224/\sqrt{4}} = 2.951 = t_{0.97,3}$$

$$\begin{aligned} 1 - \frac{\alpha}{2} &= 0.97 \\ \alpha &= 2(1 - 0.97) \\ &= 0.06 \end{aligned}$$

[sq]

More Information Unsurprisingly, if the Central Limit Theorem applies and the value of σ^2 is known, a standard normal sampling distribution may be used instead. Nevertheless, this variation will not be tested in this course.

5.5 4.2.2 Hypothesis Test For Variacnes

For a random sample of size n drawn from a normal distribution with common mean μ and common variance σ^2 ,

$$\frac{(n-1)S^2}{\sigma^2}$$

has a χ^2 distribution with $n-1$ degrees of freedom. This is the same setup for constructing a confidence interval for variances in Section 3.2.3.

As a result, a hypothesis test for σ^2 has a test statistic calculated as

$$\frac{(n-1)s^2}{\sigma^2}$$

which comes from a χ^2 sampling distribution with $n-1$ degrees of freedom.

Unlike standard normal and t , the χ^2 distribution is **not** symmetrical, and thus the critical values no longer have simplifying relationships across the three variants. If $\chi_{p,\nu}^2$ is the $100p$ th percentile of a χ^2 random variable with ν degrees of freedom, then

- For left-tailed tests, reject H_0 when

$$\frac{(n-1)s^2}{\sigma^2} \leq \chi_{\alpha,n-1}^2$$

- For two-tailed tests, reject H_0 when

$$\left[\frac{(n-1)s^2}{\sigma^2} \leq \chi_{\frac{\alpha}{2},n-1}^2 \right] \cup \left[\frac{(n-1)s^2}{\sigma^2} \geq \chi_{1-\frac{\alpha}{2},n-1}^2 \right]$$

- For right-tailed tests, reject H_0 when

$$\frac{(n-1)s^2}{\sigma^2} \geq \chi_{1-\alpha,n-1}^2$$

Example 4.2.4 SAT test scores are reported to follow a normal distribution with variance 125. A researcher questions if the assumed variance ought to be lower, and randomly samples 20 test scores. The sample variance of the 20 scores is 104.

Let $\chi_{p,\nu}^2$ be the $100p$ th percentile of a chi-squared random variable with ν degrees of freedom. The following table lists values of $\chi_{p,\nu}^2$ for specific combinations of p and ν :

	$p = 0.025$	$p = 0.05$	$p = 0.95$	$p = 0.975$
$\nu = 19$	8.907	10.117	30.144	32.852
$\nu = 20$	9.591	10.851	31.410	34.170

Test whether the researcher is justified at the 5% significance level.

Solution For σ^2 representing the variance of SAT test scores, the hypotheses are

$$H_0 : \sigma^2 = 125 H_1 : \sigma^2 < 125$$

Recall this is a left-tailed test. For $\alpha = 0.05$, the critical value is

$$\chi_{\alpha, n-1}^2 = \chi_{0.05, 19}^2 = 10.117$$

The test statistic is

$$\frac{(n-1)s^2}{\sigma^2} = \frac{19(104)}{125} = 15.808$$

Since $15.808 > 10.117$, we **fail to reject H_0 at the 5% significance level**.

[sq]

5.6 4.2.3 Contingency Table

5.6.1 Test of Goodness-of-Fit

Hypothesis tests can also assist in assessing the quality of a model. In particular, the chi-squared goodness-of-fit test checks whether a proposed distribution agrees with observed data.

Start with n independent observations that must be classified as one of r mutually exclusive categories. Define n_i as the number of observations classified as Category i , where $i = 1, 2, \dots, r$. Hence, $\frac{n_i}{n}$ is the proportion of observations in Category i .

Now consider a model that describes the distribution among the categories. If the model is properly specified, then p_i , the probability an observation belongs to Category i , should be similar to $\frac{n_i}{n}$ for all i . As such, the hypotheses can be written as

- $H_0 : p_i = \frac{n_i}{n}$ for all $i = 1, 2, \dots, r$
- $H_1 : \text{At least one } p_i \neq \frac{n_i}{n} \text{ for } i = 1, 2, \dots, r$

In other words, failing to reject H_0 suggests that the model fits the data adequately, whereas rejecting H_0 suggests that the model fits the data poorly.

Without discussing the proof, this is a right-tailed test with a test statistic calculated as

$$\sum_{i=1}^r \frac{(n_i - np_i)^2}{np_i}$$

which comes from a χ^2 sampling distribution with $r - 1$ degrees of freedom. Therefore, reject H_0 when

$$\sum_{i=1}^r \frac{(n_i - np_i)^2}{np_i} \geq \chi_{1-\alpha, r-1}^2$$

As a reminder,

- r is the number of unique categories,
- n_i is the number of Category i observations,
- n is the total number of observations,
- p_i is the model's probability of a Category i observation,
- α is the significance level, and
- $\chi_{p, \nu}^2$ is the 100 p th percentile of a χ^2 random variable with ν degrees of freedom.

Example 4.2.5 The outcomes of 150 die rolls were recorded as follows:

Die Roll	Frequency
1	17
2	18
3	24
4	29
5	33
6	29

Let $\chi_{p,\nu}^2$ be the 100p th percentile of a chi-squared random variable with ν degrees of freedom. The following table lists values of $\chi_{p,\nu}^2$ for specific combinations of p and ν :

	$p = 0.94$	$p = 0.96$	$p = 0.98$
$\nu = 5$	10.596	11.644	13.388
$\nu = 6$	12.090	13.198	15.033

Test whether the die is fair using the chi-squared goodness-of-fit test.

Solution There are six categories, one for each die roll outcome. Therefore, $r = 6$.

In addition, a fair die implies that each die roll outcome is equally likely, meaning $p_i = \frac{1}{6}$ for all i .

With 150 observations,

i	np_i
1	$150 \cdot \frac{1}{6} = 25$
2	$150 \cdot \frac{1}{6} = 25$
$\vdots$	$\vdots$
6	$150 \cdot \frac{1}{6} = 25$

The test statistic is

$$\sum_{i=1}^r \frac{(n_i - np_i)^2}{np_i} = \frac{(17-25)^2}{25} + \frac{(18-25)^2}{25} + \dots + \frac{(29-25)^2}{25} = 8.4$$

This test involves $6 - 1 = 5$ degrees of freedom. Note that

$$8.4 < 10.596$$

Determine the significance level associated with 10.596.

$$10.596 = \chi_{0.94,5}^2 = \chi_{1-\alpha,5}^2 \Rightarrow \alpha = 1 - 0.94 = 0.06$$

In conclusion, we **fail to reject H_0 at the 6% significance level** , suggesting that the assumption of a fair die seems reasonable for this data of 150 rolls.

[sq]

5.6.2 Test of Independence

A *contingency table* records the frequency of observations described by two categorical variables. It is used to examine the presence of dependence between the two variables. This is achieved using the same procedure as the goodness-of-fit test. The hypotheses are

- H_0 : The two variables are independent
- H_1 : The two variables are dependent

One variable has r number of categories, while the other variable has s . Each of the n observations belongs to one of the r -by- s combinations. Let

- n_{ij} be the number of observations in Category i for the first variable and Category j for the second variable,
- $n_{i.}$ be the subtotal number of observations in Category i for the first variable, across all categories of the second variable, and
- $n_{.j}$ be the subtotal number of observations in Category j for the second variable, across all categories of the first variable,

for $i = 1, 2, \dots, r$ and $j = 1, 2, \dots, s$. Thus, a contingency table resembles

	Second Variable		Total
	Cat1	Cat2	...
Cat1			
First			
Variable	Cat1		n_{11}
Cat2	n_{21}		n_{22}
$\vdots$	$\vdots$		$\vdots$
Cat r	n_{r1}		n_{r2}
Total	$n_{.1}$		$n_{.2}$

The test statistic is calculated as

$$\frac{1}{n} \sum_{i=1}^r \sum_{j=1}^s \frac{(n_{ij}n - n_{i.}n_{.j})^2}{n_{i.}n_{.j}}$$

which comes from a χ^2 sampling distribution with $(r-1)(s-1)$ degrees of freedom. Therefore, reject H_0 when

$$\frac{1}{n} \sum_{i=1}^r \sum_{j=1}^s \frac{(n_{ij}n - n_{i.}n_{.j})^2}{n_{i.}n_{.j}} \geq \chi_{1-\alpha, (r-1)(s-1)}^2$$

Example 4.2.6 150 vehicles were stopped at random by the police for inspection.

Type	< 2015	$\geq$ 2015	Total
Cars	40	60	100
Motorcycles	10	40	50
Total	50	100	150

Let $\chi_{p,\nu}^2$ be the 100 p th percentile of a chi-squared random variable with ν degrees of freedom. The following table lists values of $\chi_{p,\nu}^2$ for specific combinations of p and ν :

	$p = 0.95$	$p = 0.975$	$p = 0.99$
$\nu = 1$	3.841	5.024	6.635
$\nu = 2$	5.991	7.378	9.210

Test whether the vehicle type and year are independent.

Solution Note that $r = 2$ and $s = 2$, since each variable (type and year) has two categories.

The test statistic is

$$\begin{aligned}
& \frac{1}{n} \sum_{i=1}^r \sum_{j=1}^s \frac{(n_{ij}n - n_{i.}n_{.j})^2}{n_{i.}n_{.j}} \\
&= \frac{1}{150} \left(\frac{[40(150) - 100(50)]^2}{100(50)} + \frac{[60(150) - 100(100)]^2}{100(100)} \right. \\
&\quad \left. + \frac{[10(150) - 50(50)]^2}{50(50)} + \frac{[40(150) - 50(100)]^2}{50(100)} \right) \\
&= 6
\end{aligned}$$

This test involves $(2 - 1)(2 - 1) = 1$ degree of freedom. Note that

$$5.024 < 6 < 6.635$$

Determine the significance levels associated with 5.024 and 6.635.

$$\begin{aligned}
5.024 &= \chi_{0.975,1}^2 = \chi_{1-\alpha,1}^2 \Rightarrow \alpha = 1 - 0.975 = 0.025 \\
6.635 &= \chi_{0.99,1}^2 = \chi_{1-\alpha,1}^2 \Rightarrow \alpha = 1 - 0.99 = 0.01
\end{aligned}$$

In conclusion, **reject H_0 at the 2.5% significance level, but not at the 1% level**, suggesting strong evidence that vehicle type and year are dependent.

[sq]

5.7 4.2.4 Maximized Likelihoods

Recall that Examples 2.1.1. and 2.1.2 modeled the same data of 33 observations using different distributions: exponential and gamma. However, it was not addressed which distribution is a better fit. In general, such questions are difficult to answer, as it could require just as much art as science. One concept to assist us is the *maximized likelihood*. It is as it sounds – evaluate the likelihood function at the parameter values that maximize it (i.e. at the MLE estimates).

The intuition behind maximized likelihoods is simple. When a distribution produces a larger maximized likelihood, it means the data was “more likely” under that distribution. However, there are issues with comparing maximized likelihoods directly. In order to strike a balance between fitting the data and generalizing to a population, maximized likelihoods must be used intelligently.

It’s no surprise that a maximized log-likelihood is a log-likelihood function evaluated at the MLE estimates. Thus, maximized log-likelihoods are simply rescaled maximized likelihoods that carry close to the same quality of information. In fact, the methods that follow will utilize maximized log-likelihoods.

5.7.1 Likelihood Ratio Test

Consider the hypotheses

- H_0 : The data is drawn from Distribution A.
- H_1 : The data is drawn from Distribution B.

such that Distribution B is more complex than Distribution A. This means Distribution B has more free parameters than Distribution A. A free parameter is simply a parameter whose value is not specified. In addition, Distribution A must be a special case of Distribution B. Examples include:

- Exponential is a special case of gamma (with shape parameter of 1).
- A distribution with pre-specified parameter values is a special case of the same distribution with free parameters.

In short, we prefer to use a simpler model (i.e. fewer free parameters), but will defer to a more complex model (i.e. more free parameters) if there is significant improvement.

Define

- r_0 as the number of free parameters in Distribution A,
- r_1 as the number of free parameters in Distribution B,
- l_0 as the maximized log-likelihood under H_0 , and
- l_1 as the maximized log-likelihood under H_1 .

Under appropriate conditions, this right-tailed test has a test statistic calculated as

$$2(l_1 - l_0)$$

which comes from a χ^2 sampling distribution with $r_1 - r_0$ degrees of freedom. Therefore, reject H_0 when

$$2(l_1 - l_0) \geq \chi_{1-\alpha, r_1-r_0}^2$$

Clearly, the likelihood ratio test favors H_1 only if l_1 is sufficiently larger than l_0 .

Example 4.2.7 Scientists have noticed an unusual mortality rate for a certain breed of dogs. They considered two distributions to model the data. Their findings are as follows:

- For an exponential distribution with 1 free parameter, the likelihood function was maximized at $2.1135 \cdot 10^{-11}$.
- For a Weibull distribution with 2 free parameters, the likelihood function was maximized at $2.6385 \cdot 10^{-11}$.

Let $\chi_{p,\nu}^2$ be the 100 p th percentile of a chi-squared random variable with ν degrees of freedom. The following table lists values of $\chi_{p,\nu}^2$ for specific combinations of p and ν :

	$p = 0.90$	$p = 0.95$	$p = 0.975$
$\nu = 1$	2.706	3.841	5.024
$\nu = 2$	4.605	5.991	7.378

Determine the result of the likelihood ratio test.

Solution Since the exponential distribution has fewer free parameters, it is understood to be the distribution supported by the null hypothesis. Thus, $r_0 = 1$ and $r_1 = 2$.

The test statistic is

$$\begin{aligned}
2(l_1 - l_0) &= 2(\ln[L_1] - \ln[L_0]) \\
&= 2(\ln[2.6385 \cdot 10^{-11}] - \ln[2.1135 \cdot 10^{-11}]) \\
&= 0.444
\end{aligned}$$

This test involves $2 - 1 = 1$ degree of freedom. Note that

$$0.444 < 2.706$$

Determine the significance level associated with 2.706.

$$2.706 = \chi_{0.9,1}^2 = \chi_{1-\alpha,1}^2 \Rightarrow \alpha = 1 - 0.9 = 0.1$$

In conclusion, we **fail to reject H_0 at the 10% significance level**, suggesting that the exponential distribution is preferred over the Weibull distribution. The additional one free parameter from Weibull did not produce a significantly better fit.

[sq]

5.7.2 Information Criteria

There are other approaches to model selection besides hypothesis testing, one of which uses information criteria. In this course, we will only discuss the Bayesian information criterion (BIC). The idea is to calculate a BIC value for each model of interest. The model with the largest BIC value is judged to be the best. The value is calculated as

$$l - \frac{r}{2} \ln n$$

where

- l is the maximized log-likelihood,
- r is the number of free parameters, and
- n is the sample size.

Keep in mind that free parameters are also parameters to be estimated via MLE. In addition, the formula shows that models with higher complexity (i.e. more free parameters) are penalized more. This allows models of various degrees of complexity to be compared on an equal footing.

Example 4.2.8 You consider three models for fitting a particular set of 300 observations. You are given:

Model	Number of Estimated Parameters	Maximized Log-Likelihood
I	3	-610
II	4	-608
III	5	-605

Select the best model based on the Bayesian information criterion.

Solution

Calculate the BIC value for each model. For Model I, it is

$$-610 - \frac{3}{2} \ln 300 = -618.5557$$

The value for each model is tabulated below:

Model	r	l	BIC
I	3	-610	-618.5557
II	4	-608	-619.4076
III	5	-605	-619.2595

Model I has the largest value.

[sq]

5.8 4.2.5 Neyman Pearson Lemma

Finally, consider the following set of hypotheses involving a generic parameter θ and two constants d_0 and d_1 :

$$H_0 : \theta = d_0 \quad H_1 : \theta = d_1$$

According to the Neyman-Pearson lemma, we will obtain the *most powerful test* with significance level α when the rejection region is derived from

$$\frac{L(d_0)}{L(d_1)} \leq r$$

where $L(\cdot)$ is the likelihood function. Said differently, among rejection regions with the same α , the region derived from this inequality will have the largest power.

It helps to remember that a likelihood function is also a function of the observed values $x_1, x_2, \dots, x_n$. Likewise, the ratio of likelihood functions is a function of the observed values as well. As a result, the inequality can be manipulated until a meaningful test statistic emerges, such as $\bar{x}$ for example. Doing so helps to clarify the rejection region.

The constant r is similar to a critical value. But since the inequality is to be manipulated, the value of r is likely unimportant. In any case, it is clear that the inequality captures the idea of rejecting H_0 when $L(d_1)$ significantly exceeds $L(d_0)$.

To illustrate this process, assume that we have a distribution with the following probability function:

$$f(x) = \frac{1}{\theta} e^{-x/\theta}, x > 0$$

We want to test the following hypotheses regarding the mean parameter of the distribution.

- $H_0 : \theta = 10$
- $H_1 : \theta = 12$

Let's say we randomly draw a single observation from the distribution. Intuitively, a reasonable hypothesis test should reject H_0 in favor of H_1 if the observation is **large** enough, as larger observations are more likely with a larger mean parameter (i.e., 10 vs. 12). In technical terms, this intuition describes a right-tailed test where the null hypothesis is rejected when the observation exceeds a certain threshold.

To verify that a right tail rejection region is reasonable, we can use the Neyman-Pearson lemma:

$$\frac{L(10)}{L(12)} \leq r$$

As we are only drawing one observation:

$$\frac{L(10)}{L(12)} = \frac{\frac{1}{10}e^{-x/10}}{\frac{1}{12}e^{-x/12}} = \frac{6}{5}e^{-x/60}$$

This means the rejection region is defined by:

$$\begin{aligned} \frac{6}{5}e^{-x/60} &\leq r \\ e^{-x/60} &\leq \frac{5}{6}r \\ -\frac{x}{60} &\leq \ln\left(\frac{5}{6}r\right) \\ x &\geq -60\ln\left(\frac{5}{6}r\right) \\ x &\geq k \end{aligned}$$

where k is a constant.

In other words, to achieve the most powerful test, we would reject the null hypothesis when the observation is at least a specific constant k . This finding aligns with our initial suspicion of a right-tailed test being reasonable.

Coach's Remarks

Notice how we simplify the last two lines of the inequality above by assuming the following:

$$k = -60\ln\left(\frac{5}{6}r\right)$$

This substitution is done because the exact expression in terms of r will likely be inconsequential. By using k instead of the complex expression in terms of r , the final line allows us to focus better in describing the rejection region sensibly (i.e., x exceeds a constant) rather than having an inequality that does not appear as helpful.

If we assume a significance level of 5%, that means there is a 5% chance of rejecting the null hypothesis when it is true:

$$Pr(X \geq k \mid \theta = 10) = 0.05$$

With the density function of X and a mean parameter of 10, we can solve for the value of k :

$$Pr(X \geq k \mid \theta = 10) = \int_k^{\infty} \frac{1}{10}e^{-x/10}dx = e^{-k/10}$$

$$e^{-k/10} = 0.05$$

$$k = 29.9573$$

In conclusion, the most powerful rejection region in this scenario is $x \geq 29.9573$.

Coach's Remarks

If you are interested, you can solve for the value of r based on the result of $k = 29.9573$ given that $k = -60 \ln \left(\frac{5}{6} r \right)$. This produces $r = 0.72835$.

That gives us the following rejection region in the unit of the likelihood ratio:

$$\frac{L(10)}{L(12)} \leq 0.72835$$

Compare this to the rejection region expressed using k :

$$x \geq 29.9573$$

Both inequalities describe the same information. However, notice how the second form is considerably more practical than the first. The observation can be directly compared with $k = 29.9573$ to determine if the null hypothesis is rejected. In contrast, the first form requires working through complex algebra to arrive at the test conclusion. This illustrates how r is merely a constant that is relatively unimportant once we have derived the inequality form that directly involves the observations (in other words, when a meaningful test statistic emerges).

With a critical value of $k = 29.9573$, we can also solve for the power of the test (i.e., the probability of rejecting the null hypothesis when it is false):

$$Pr(X \geq 29.9573 \mid \theta = 12) = e^{-29.9573/12} = 0.08238$$

By the lemma, this means that with a significance level set at 0.05, the test can reach a power of at most 0.08238. All other tests (with the same α) will not possess a larger power.

It is worth reiterating that generally x is not the test statistic that will emerge from manipulating the inequality. Depending on the likelihood function, the test statistic could be $\bar{x}$, $\sum x_i$, $\prod x_i^2$, and so on, just to name a few. The key is that we generally aim to structure the test statistic in a meaningful way that isolates the observations on one side of the inequality.

Example 4.2.9 A certain brand of LED lightbulbs have exponentially distributed lifetimes with PDF

$$f(x) = \frac{1}{\theta} e^{-x/\theta}, x > 0$$

A random sample of 15 lightbulbs is used to test following set of hypotheses:

- $H_0 : \theta = 4$
- $H_1 : \theta = 3$

Determine the rejection region for the most powerful test.

Solution: First, determine the likelihood function.

$$\begin{aligned} L(\theta) &= \prod_{i=1}^{15} f(x_i) \\ &= \frac{1}{\theta^{15}} e^{-(\sum_{i=1}^{15} x_i)/\theta} \end{aligned}$$

Therefore,

$$\begin{aligned} \frac{L(4)}{L(3)} &= \frac{\frac{1}{4^{15}} e^{-(\sum_{i=1}^{15} x_i)/4}}{\frac{1}{3^{15}} e^{-(\sum_{i=1}^{15} x_i)/3}} \\ &= \left(\frac{3}{4} \right)^{15} e^{(\sum_{i=1}^{15} x_i)/12} \end{aligned}$$

Applying the lemma,

$$\begin{aligned}
& \left(\frac{3}{4}\right)^{15} e^{(\sum_{i=1}^{15} x_i)/12} \leq r \\
\Rightarrow & e^{(\sum_{i=1}^{15} x_i)/12} \leq r \left(\frac{4}{3}\right)^{15} \\
\Rightarrow & \frac{\sum_{i=1}^{15} x_i}{12} \leq \ln \left[r \left(\frac{4}{3}\right)^{15} \right] \\
\Rightarrow & \sum_{i=1}^{15} x_i \leq 12 \ln \left[r \left(\frac{4}{3}\right)^{15} \right]
\end{aligned}$$

Manipulating the inequality resulted in a summation test statistic emerging on the left side of inequality. Thus, the right side of the inequality is the critical value in the unit of total lifetime.

The most powerful rejection region has the form

$$\sum_{i=1}^{15} x_i \leq c$$

This makes sense; if the sum of the observations is sufficiently small, then a smaller exponential mean seems more reasonable than a larger one, thus favoring H_1 over H_0 .

[sq]

Coach's Remarks A significance level is needed to compute the critical value c . On the other hand, the sampling distribution is gamma with shape parameter 15 and scale parameter θ . This makes it challenging to solve by hand, and hence is beyond the scope of this course. Nevertheless,

$$Pr \left(\sum_{i=1}^{15} X_i \leq 36.985 | \theta = 4 \right) = 0.05$$

meaning $c = 36.985$ at the 5% significance level. In turn, the power is

$$Pr \left(\sum_{i=1}^{15} X_i \leq 36.985 | \theta = 3 \right) = 0.2585$$

Therefore, the Neyman-Pearson lemma guarantees that all other $\alpha = 0.05$ tests will have a power no greater than 0.2585.

Example 4.2.10

Suppose 6 observations are drawn from a population with the following PDF:

$$f(x) = \left(\frac{1}{\theta}\right) 3x^2 e^{-x^3/\theta}, x > 0$$

A hypothesis test is performed to test the following hypotheses:

- $H_0 : \theta = 3$
- $H_1 : \theta = 7$

Using the Neyman-Pearson lemma, determine the form of the best rejection region.

Solution:

Solve for the general form of the likelihood function with n observations:

$$\begin{aligned} L(\theta) &= \prod_{i=1}^n \left[\left(\frac{1}{\theta}\right) 3x_i^2 e^{-x_i^3/\theta} \right] \\ &= \left(\frac{1}{\theta}\right)^n \cdot 3^n \left(\prod_{i=1}^n x_i\right)^2 \cdot e^{-(\sum_{i=1}^n x_i^3)/\theta} \end{aligned}$$

Based on each of the hypotheses, substitute the value of θ and $n = 6$ into the likelihood function:

$$L(\theta = 3) = \left(\frac{1}{3}\right)^6 \cdot 3^6 \left(\prod_{i=1}^6 x_i\right)^2 \cdot e^{-(\sum_{i=1}^6 x_i^3)/3}$$

$$L(\theta = 7) = \left(\frac{1}{7}\right)^6 \cdot 3^6 \left(\prod_{i=1}^6 x_i\right)^2 \cdot e^{-(\sum_{i=1}^6 x_i^3)/7}$$

Therefore,

$$\begin{aligned} \frac{L(\theta=3)}{L(\theta=7)} &= \frac{\left(\frac{1}{3}\right)^6 \cdot 3^6 \left(\prod_{i=1}^6 x_i\right)^2 \cdot e^{-(\sum_{i=1}^6 x_i^3)/3}}{\left(\frac{1}{7}\right)^6 \cdot 3^6 \left(\prod_{i=1}^6 x_i\right)^2 \cdot e^{-(\sum_{i=1}^6 x_i^3)/7}} \\ &= \left(\frac{7}{3}\right)^6 e^{(1/7-1/3)\sum_{i=1}^6 x_i^3} \end{aligned}$$

Applying the lemma:

$$\begin{aligned} \left(\frac{7}{3}\right)^6 e^{(1/7-1/3)\sum_{i=1}^6 x_i^3} &\leq r \\ e^{(1/7-1/3)\sum_{i=1}^6 x_i^3} &\leq \left(\frac{3}{7}\right)^6 r \\ \left(\frac{1}{7} - \frac{1}{3}\right) \sum_{i=1}^6 x_i^3 &\leq \ln \left[\left(\frac{3}{7}\right)^6 r \right] \\ \sum_{i=1}^6 x_i^3 &\geq \frac{\ln \left[\left(\frac{3}{7}\right)^6 r \right]}{\frac{1}{7} - \frac{1}{3}} \end{aligned}$$

The last inequality sign is flipped because $\frac{1}{7} - \frac{1}{3}$ is a negative number. Therefore, the best rejection region is defined by:

$$\sum_{i=1}^6 x_i^3 \geq c$$

[sq]

5.9 4.2 Final Exam Tips

You've made it to the end of Learn Mathematical Statistics! Be sure to schedule your final exam by emailing proctoring@coachingactuaries.com if you haven't already.

Please note that most (if not all) of the final exam questions will be ones you have not seen on the assignments. Reviewing and retaking the assignments will still be helpful in preparing for the final exam, but it is more important to have a good understanding of the concepts covered in Learn.

5.10 4.2 Overview

In the previous subsection, the focus was on learning the methodology and key terminology used in hypothesis tests. In this subsection, we graduate from the trivial setup that had only one observation, and now consider setups of more substance.

We begin by looking at hypothesis tests for parameters, specifically mean and variance. Next, we will discuss how hypothesis tests can be used in model assessment or selection. Finally, the Neyman-Pearson lemma will be presented.
